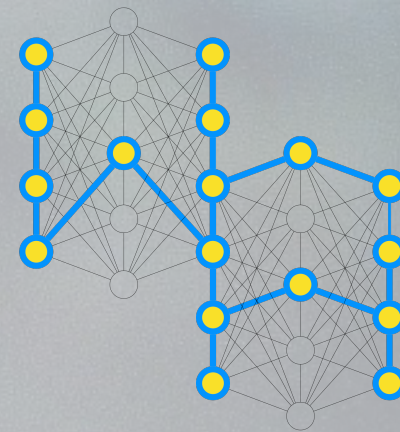


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ANNI



UNIVERSITÀ
DEGLI STUDI
DI PADOVA



Introduction to Webapps

Web Applications

Master Degree in Computer Engineering

Master Degree in Cybersecurity

Master Degree in ICT for Internet and Multimedia

Academic Year 2023/2024



DIPARTIMENTO
DI INGEGNERIA
DELL'INFORMAZIONE

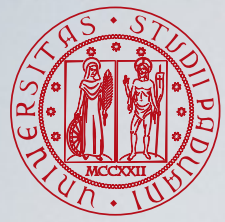
Nicola Ferro

Intelligent Interactive Information Access (IIIA) Hub

Department of Information Engineering

University of Padua



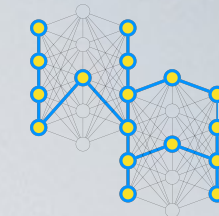
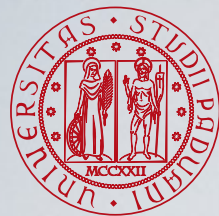


Outline

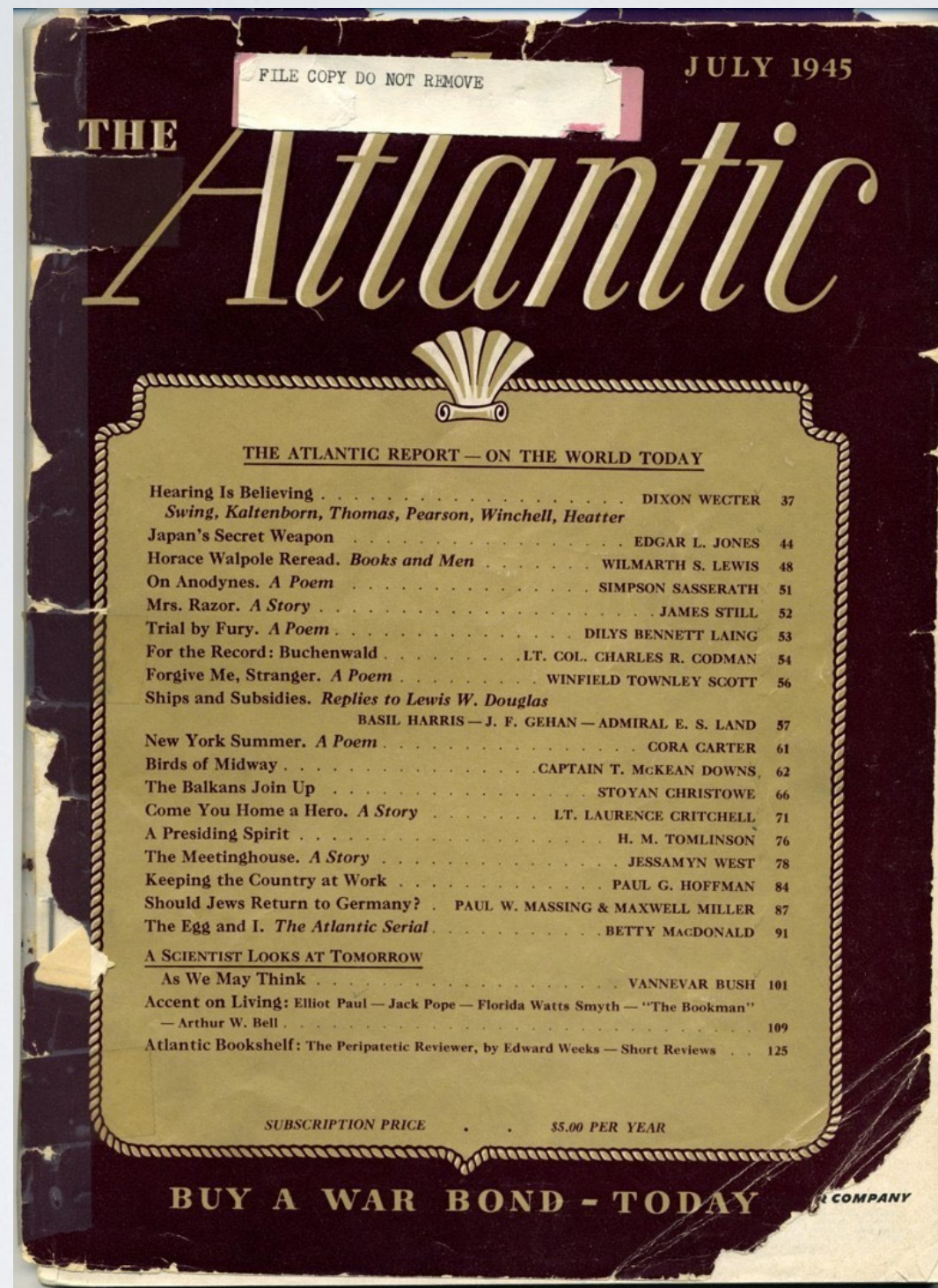


- A bit of history
- Recap on distributed applications

A bit of history



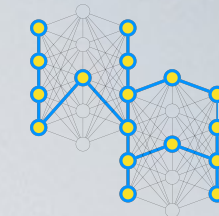
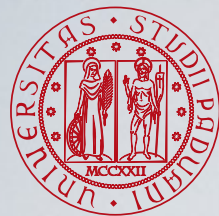
Vannevar Bush: The Memex (July 1945)



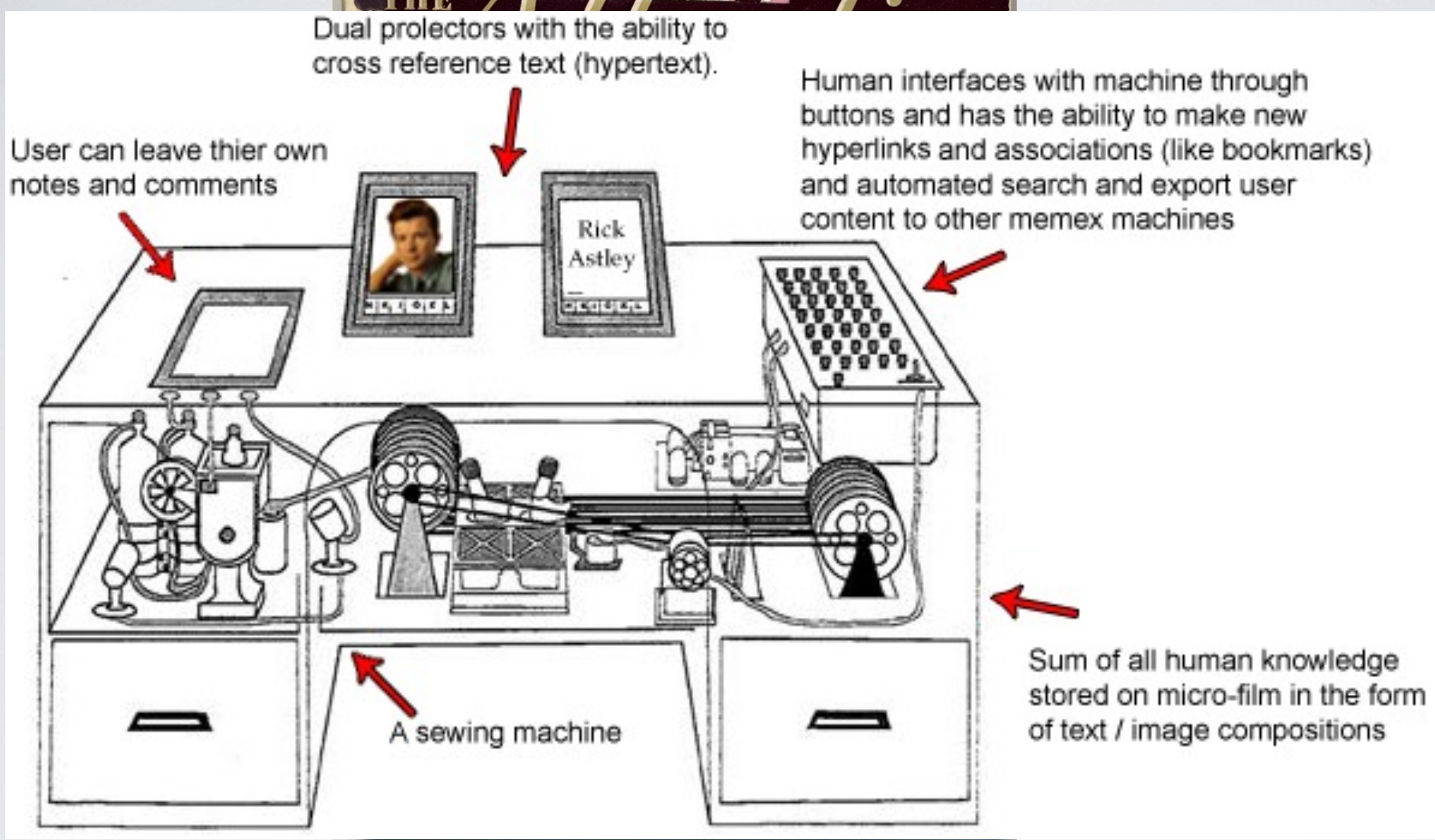
Vannevar Bush (1890–1974)
Director of Office of Scientific
Research and Development (OSRD)



Bush, V. (1945). As We May Think. *The Atlantic Monthly*, 176:101–108.
<https://www.theatlantic.com/magazine/archive/1945/07/as-we-may-think/303881/>



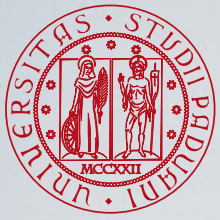
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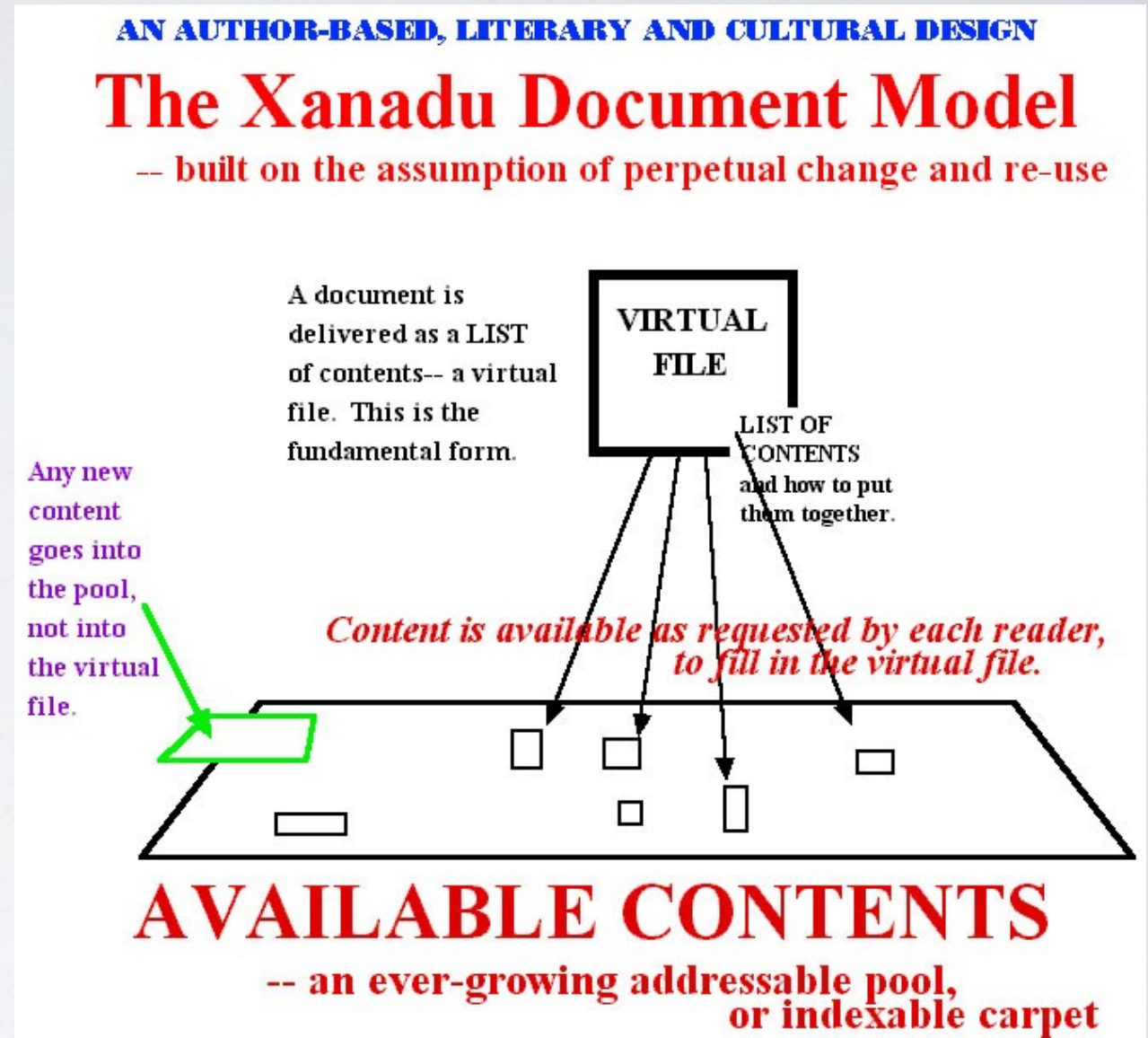
Bush, V. (1945). As We May Think. *The Atlantic Monthly*, 176:101–108.
<https://www.theatlantic.com/magazine/archive/1945/07/as-we-may-think/303881/>



Ted Nelson: Hypertext (1963/1965)

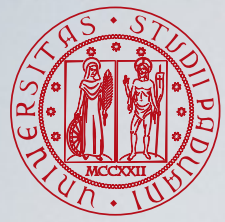


- Theodor Holm Nelson (born 1937) is a philosopher and sociologist, pioneer of ICT
- He coined the term hypertext in 1963, first published in print in an interview at Vassar College, NY, in 1965
 - “He introduced the concept of the **hyper-text**, which would be a more flexible, more generalized, **non-linear** presentation of material on a particular subject.”
- Project Xanadu (1960—1998), first hypertext project, released as open source in 2014
 - transcopyright, original context, bi-directional links, versioning, ...



Nelson, T. H. (1999). Xanalogical Structure, Needed Now More than Ever: Parallel Documents, Deep Links to Content, Deep Versioning, and Deep Re-Use. *ACM Computing Surveys (CSUR)*, 31(4es):33:1–33:32.

Wedeles, L. (1965). Prof. Nelson Talk Analyzes P.R.I.D.E. *Vassar Miscellany News*.
http://facultysites.vassar.edu/mijoyce/MiscNews_Feb65.html



Douglas Engelbart: NLS System (1962-1968)

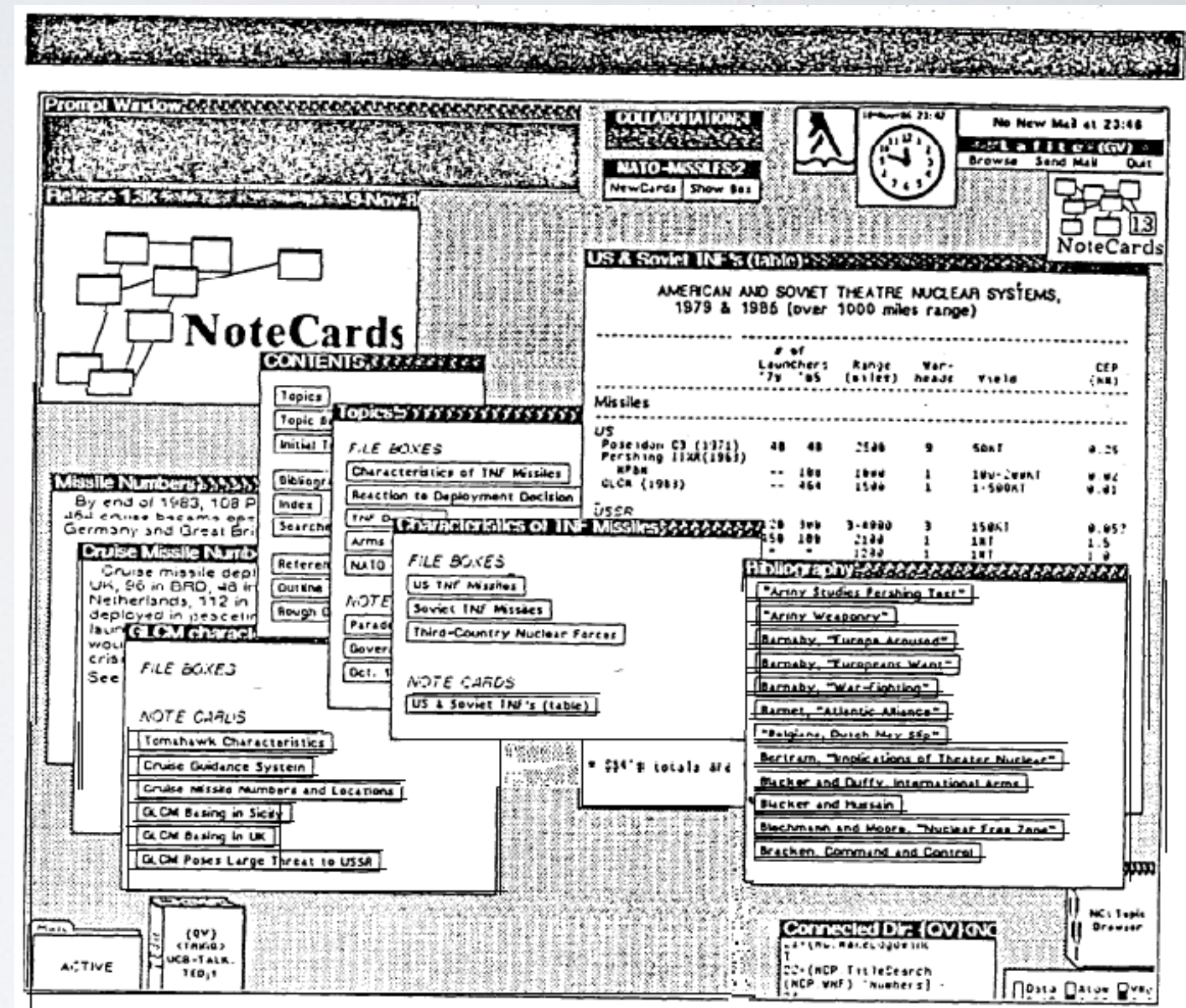
- Douglas Carl Engelbart (1925—2013), Augmentation Research Center Lab at Stanford Research Institute
- The oN-Line System (NLS) was the first to use mouse, hyperlinks, windows, presentation programs, ...
- the “Mother of All Demos” at the ACM/IEEE Fall Joint Computer Conference in 1968



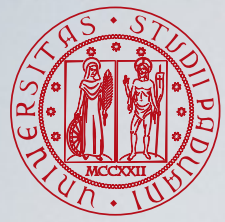
Engelbart, D. C. and English, W. K. (1968). A research center for augmenting human intellect. *In Proc. December 9-11, 1968, Fall Joint Computer Conference, (AFIPS 1968, Fall, Part I)*, pages 395–410. The Thompson Book Company, Washington, USA.

Halasz/Moran/Trigg: Notecards (1983-1987)

- NoteCards was developed by Xerox in LISP and was one of the most successful/known hypertext systems
- NoteCards goal was to support analysts in collecting and organising information in order to prepare summary reports
- NoteCards was extensible via a programming language based on LISP



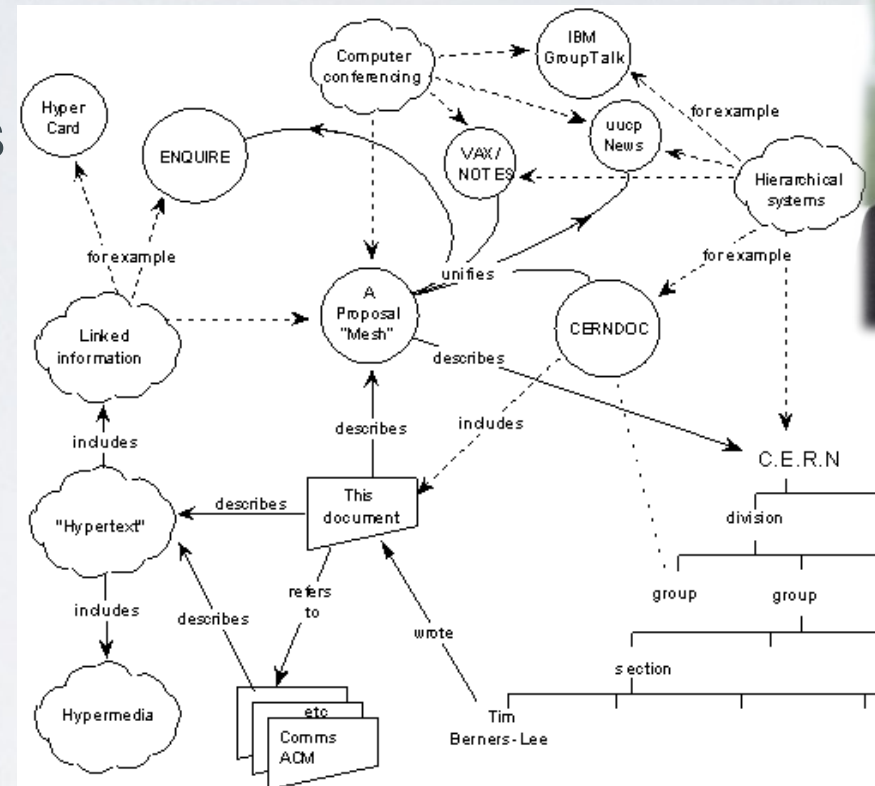
Halasz, F. G., Moran, T. P., and Trigg, R. H. (1987). Notecards in a Nutshell. In Carroll, J. M. and Tanner, P. P., editors, *Proc. Conference on Human Factors in Computing Systems and Graphics Interface (CHI 1987)*, pages 45–52. ACM Press, New York, USA.



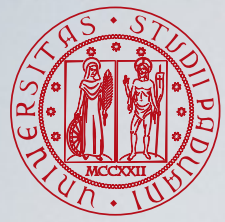
Tim Berners-Lee: World Wide Web (1989-1991)



- Tim Berners-Lee proposed (in 1989 and again in 1990) the Web as an linked information system for managing outputs from large projects and experiments at CERN
- The Web simplified previous research on hypertext/hypermedia systems and was a perfect “marriage” with the Internet
- During 1990 internal development at CERN
- 1991 release of the first Web site at CERN (<http://info.cern.ch/>)
- Poster about WWW accepted at the 3rd ACM Conference on Hypertext in 1991



Berners-Lee, T. (1989). Information Management: A Proposal. CERN, Switzerland
<https://www.w3.org/History/1989/proposal.html>



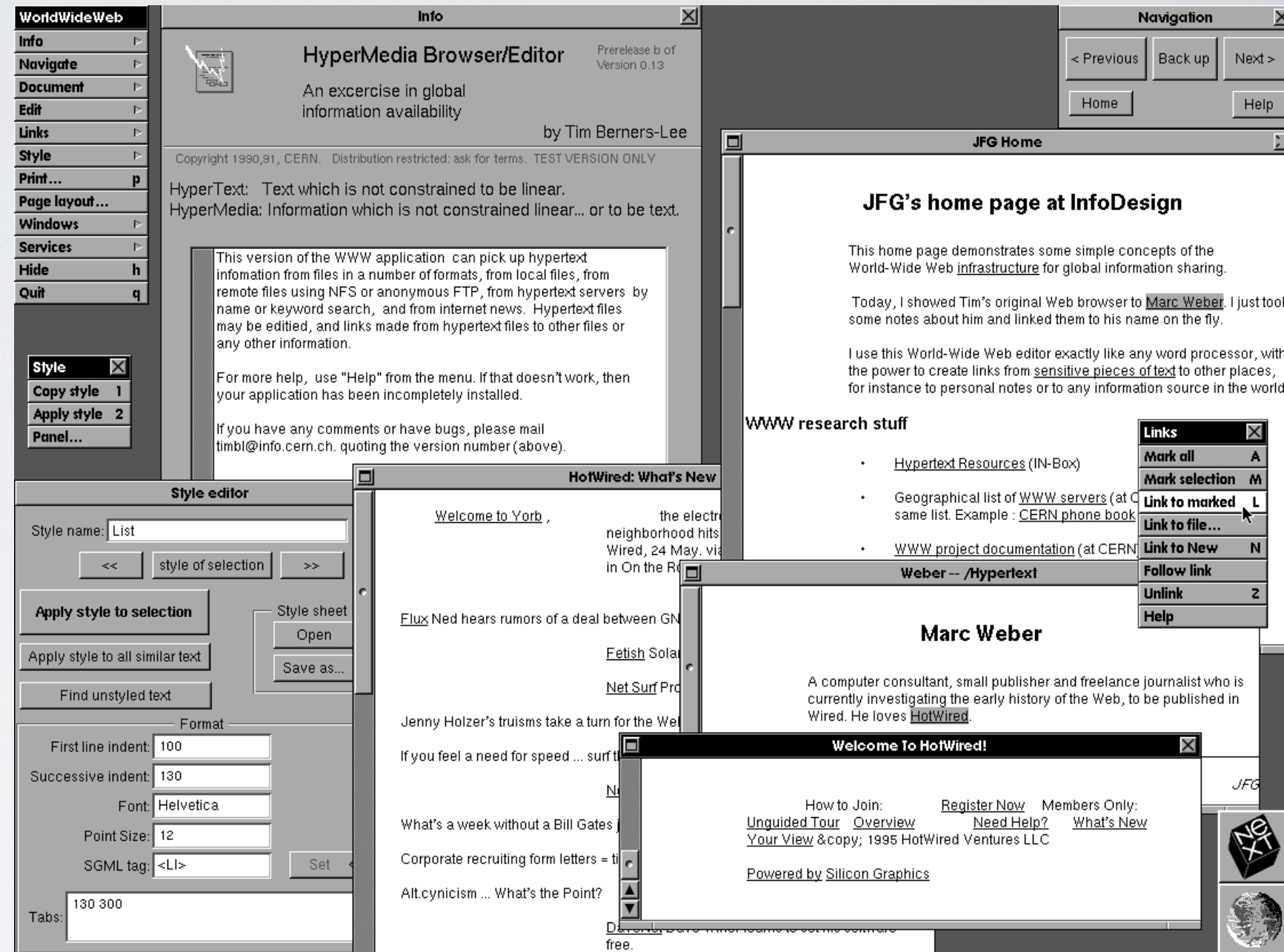
Browsers: Early Days

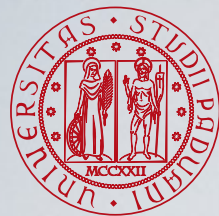


Browsers: Early Days

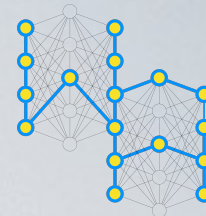


- 1990-1991: Tim Berners-Lee/
CERN, **WorldWideWeb/Nexus**
WYSIWIG browser/editor on
NeXT





Browsers: Early Days



- 1990-1991: Tim Berners-Lee/
CERN, **WorldWideWeb/Nexus**
WYSWIG browser/editor on
NeXT
- 1991-1992: Nicola Pellow/
CERN, **Line Mode Browser**,
only textual, available for many
platforms

```
CERN Welcome

CERN

The European Laboratory for Particle Physics, located near Geneva[1] in
Switzerland[2] and France[3]. Also the birthplace of the World-Wide
Web[4].

This is the CERN laboratory main server. The support team provides a set of
Services[5] to the physics experiments and the lab. For questions and
suggestions, see WWW Support Contacts[6] at CERN

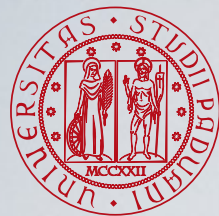
-----
About the Laboratory[7] - Hot News[8] - Activities[9] - About Physics[10] -
Other Subjects[11] - Search[12]
-----

About the Laboratory

Help[13] and General information[14], divisions, groups and
activities[15] (structure), Scientific committees[16]

Directories[17] (phone & email, services & people), Scientific
Information Service[18] (library, archives or Alice), Preprint[19] Server

1-45, Back, Up, <RETURN> for more, Quit, or Help: █
```

Browsers: Early Days



- 1990-1991: Tim Berners-Lee/
CERN, **WorldWideWeb/Nexus**
WYSIWIG browser/editor on
NeXT
- 1991-1992: Nicola Pellow/
CERN, **Line Mode Browser**,
only textual, available for many
platforms
- 1992: University of Kansas,
Lynx textual browser, originally
for Unix-like platforms, now
multiplatform, still supported,
some navigational features

```
Lynx Information

Lynx

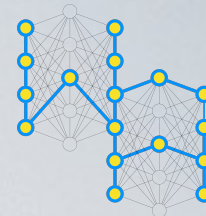
Lynx is a text browser for the World Wide Web.      Lynx 2.8.3 runs on
Un*x, VMS, Windows 95/98/NT but not 3.1 or 3.11, on DOS (386 or
higher) and OS/2 EMX.  The current developmental version is also
available for testing.  Ports to Mac are in beta test.

* Many user questions are answered in the online help provided with
Lynx. Press the '?' key to find this help.
* If you are encountering difficulty with Lynx you may write to
lynx-dev@sig.net. Be as detailed as you can about the URL where
you were on the Web when you had trouble, what you did, what Lynx
version you have (try '=' key), and what OS you have. If you are
using an older version, you may well need to upgrade.

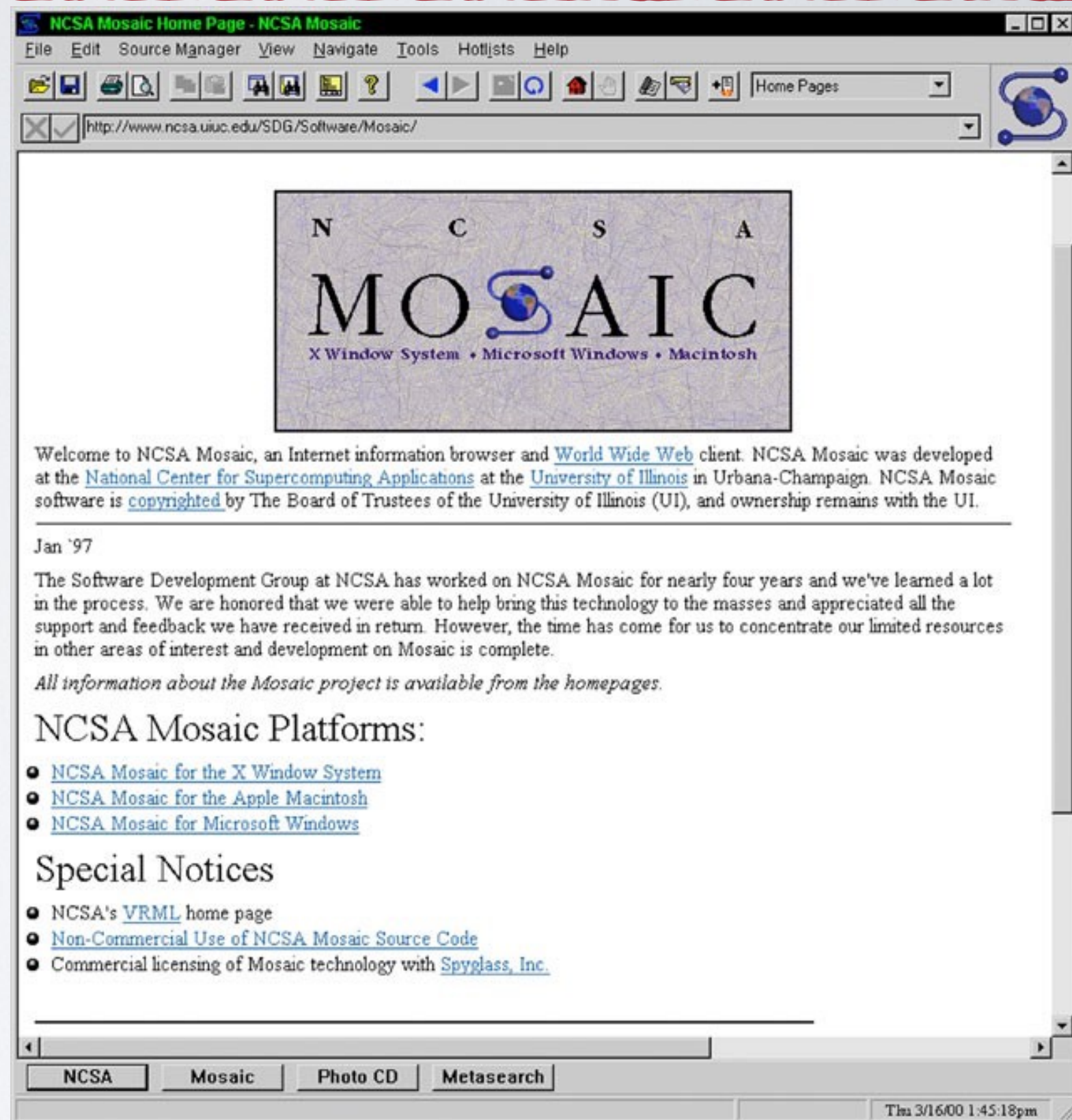
Maintained by lynxdev@browser.org.

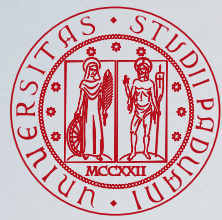
Commands: Use arrow keys to move, '?' for help, 'q' to quit, '<-' to go back.
Arrow keys: Up and Down to move. Right to follow a link; Left to go back.
H)elp O)ptions P)rint G)o M)ain screen Q)uit /=search [delete]=history list
```


Browsers: Early Days



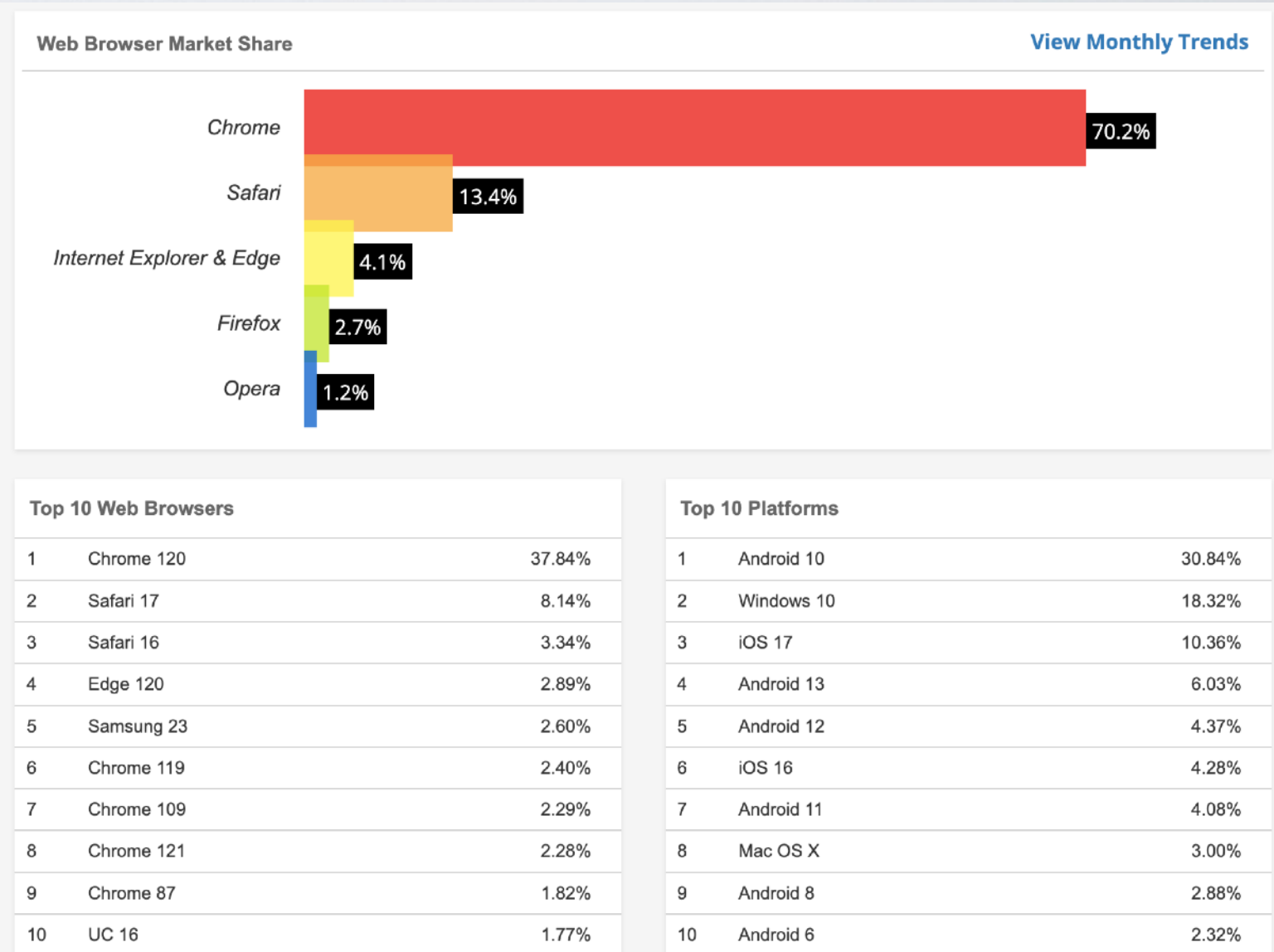
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CERN, **Line Mode Browser**,
only textual, available for many
platforms
- 1992: University of Kansas,
Lynx textual browser, originally
for Unix-like platforms, now
multiplatform, still supported,
some navigational features
- 1993: University of Illinois
Urbana-Champaign, **Mosaic**
graphical browser on multiple
platforms, popularised the Web





Browsers: Closer to Us

- 1994 Netscape Navigator
- 1995 Microsoft Internet Explorer
- 1996 Opera
- 1998 Netscape launched the Mozilla Foundation to develop an open source browser
- 2003 Safari
- 2004 Firefox
 - supersedes Netscape Navigator
- 2008 Google Chrome
- 2015 Microsoft Edge
 - supersedes Internet Explorer



Browser Market Shares for January 2024 from <https://www.w3counter.com/globalstats.php>

Web 1.0: “Read Web” - Informative Web

- Roughly 1990—2000
- Technologies
 - HTTP
 - HTML
 - MIME
 - URL
- Static Web Pages
- Dynamic Web Pages
- Web Portals



Web 2.0: “Read/Write Web” - Participative Web

- Roughly 2000—2010 (still on)

- Technologies

- XML

- AJAX

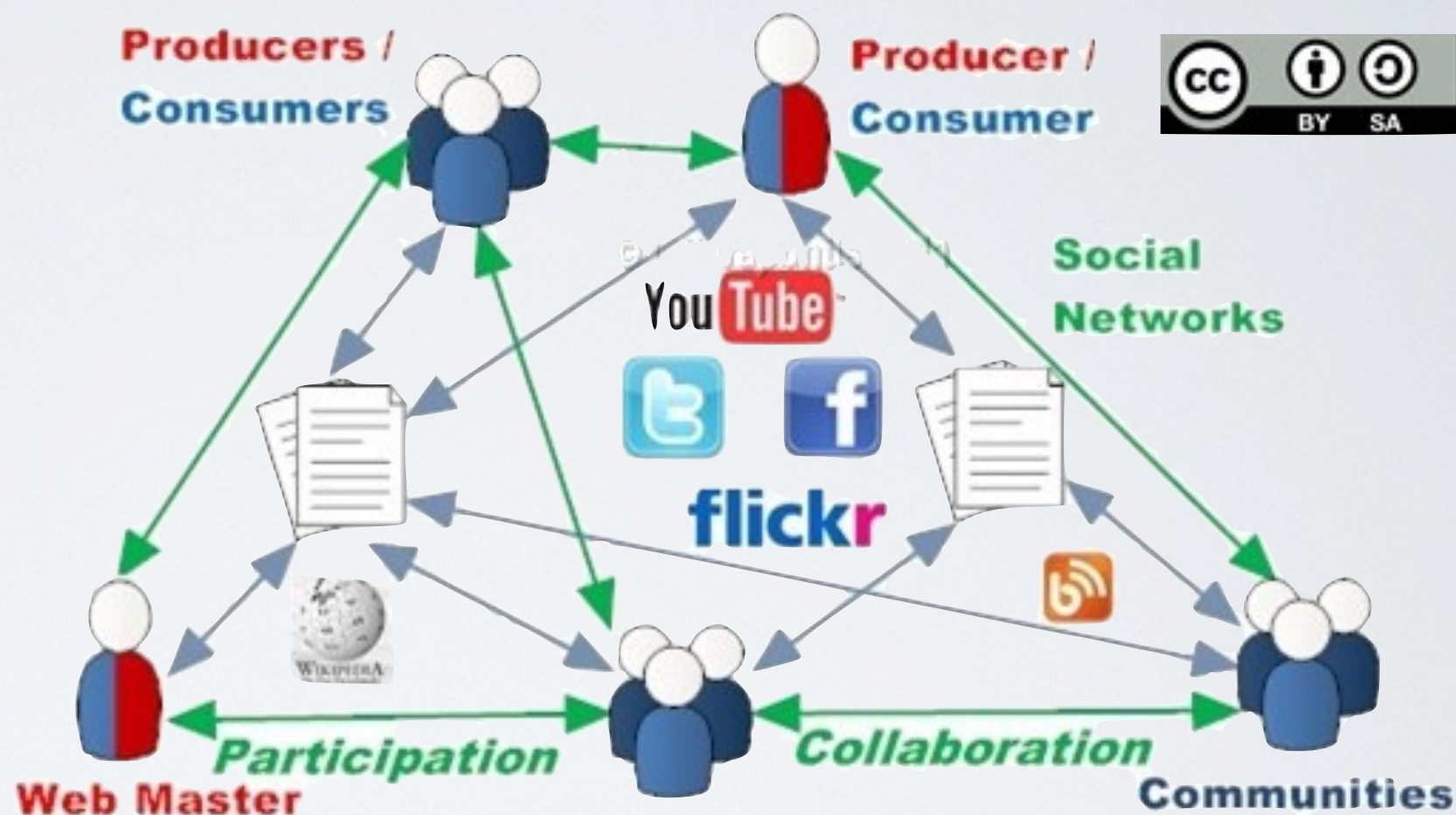
- JSON

- Social Media

- Web Services

- SOAP

- REST



Web 3.0: “Web of Data” / “Semantic Web” - Intelligent Web

Roughly 2010—onwards

supposed to be Web 2.0

Technologies

RDF

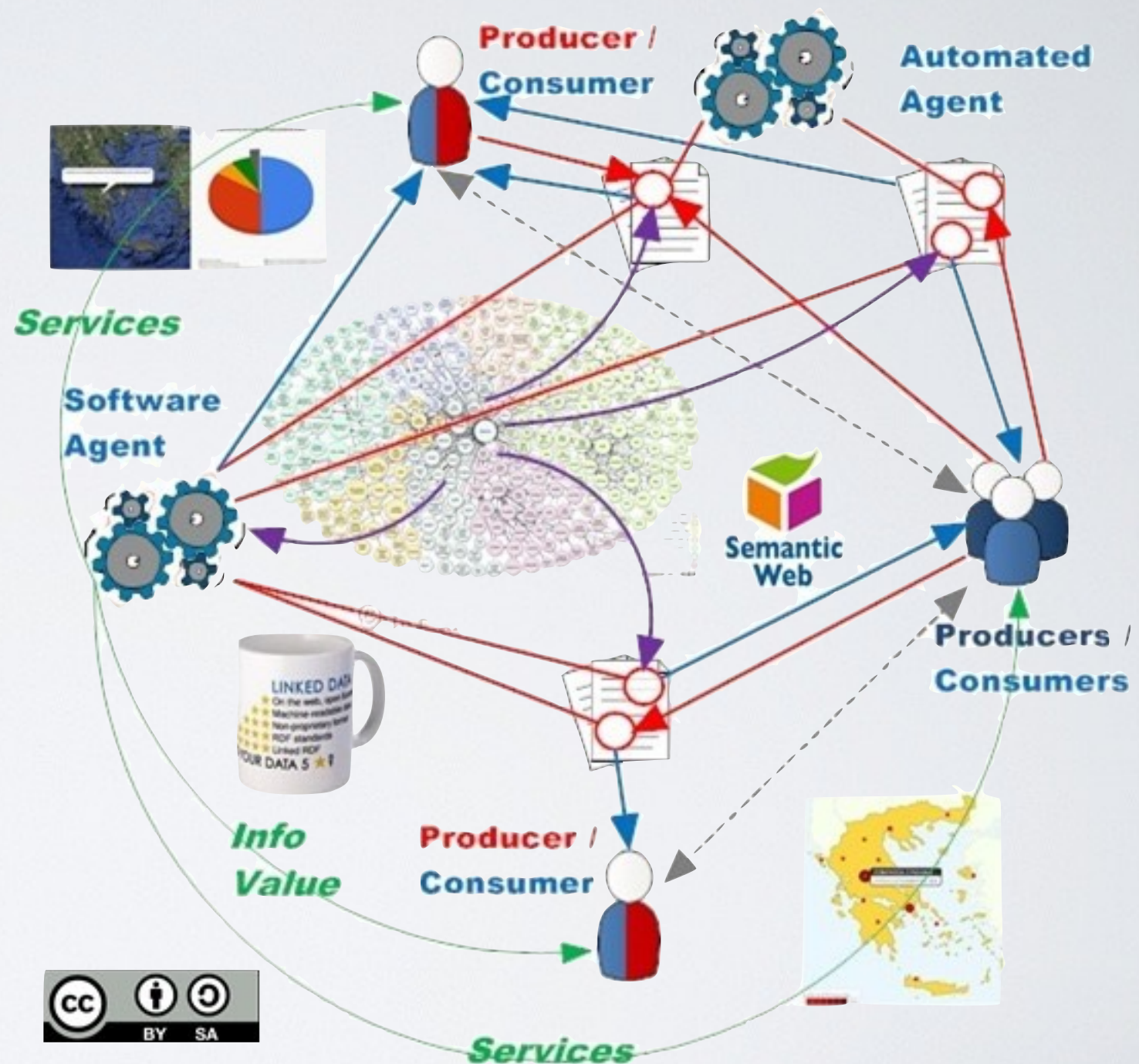
OWL

SPARQL

Machine processable/
executable information

Linked (Open) Data

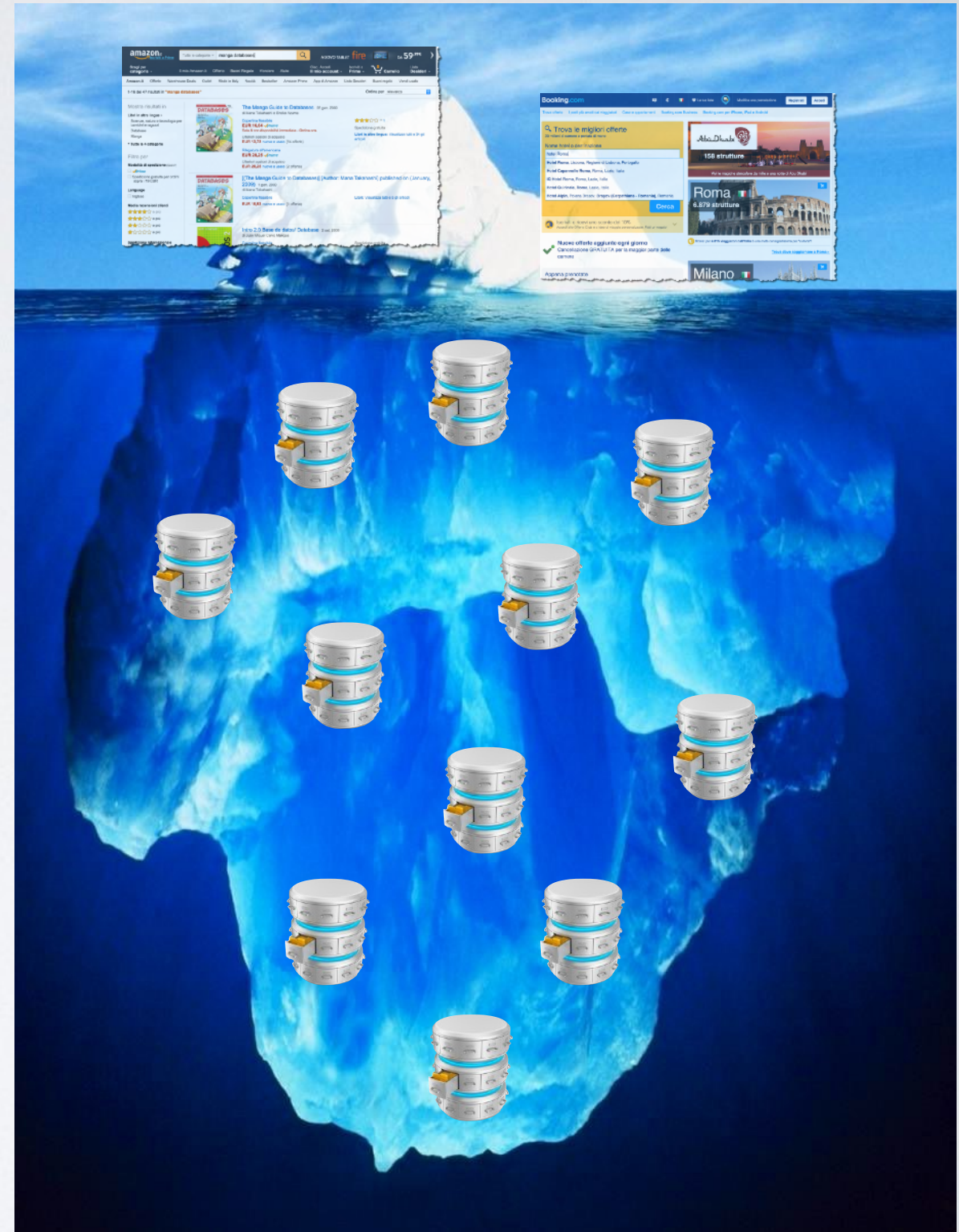
Big Data



The Deep Web

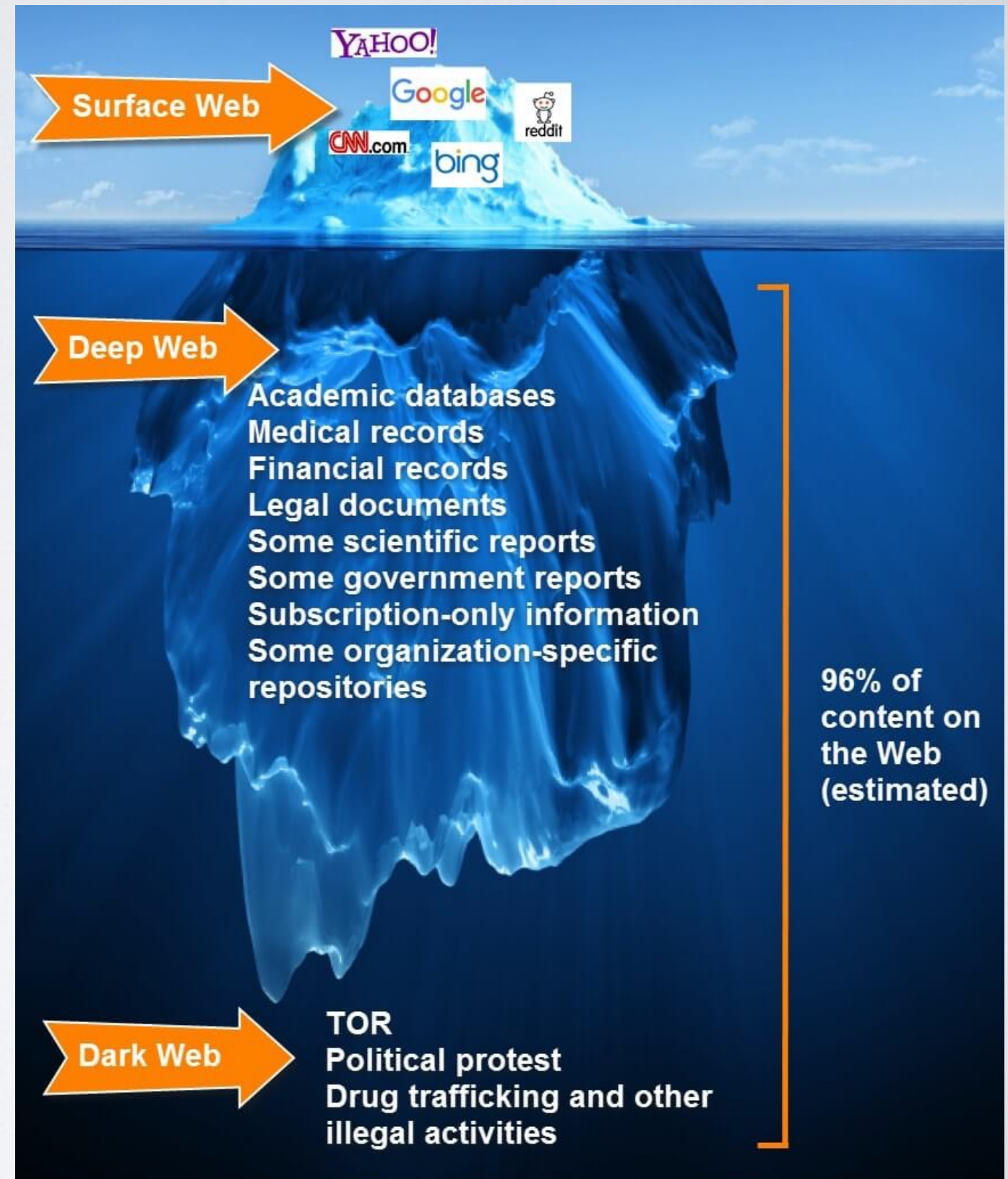


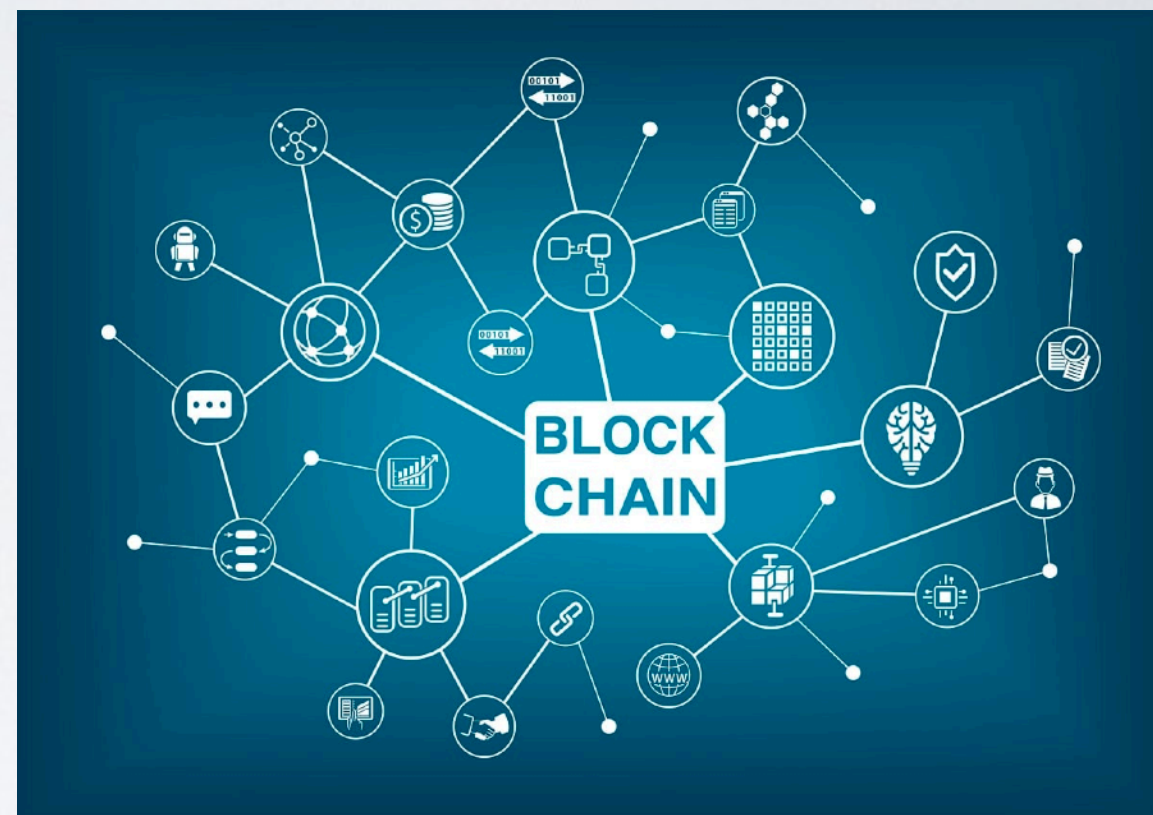
- Raw figures estimate the existence of about **14 billion raw tables**, coming from **5.4 million schemas** and comprising more than **5.4 million attributes**
 - Gartner puts the **relational database market** at \$26 billion with about **9% annual growth** for an expected \$40 billion market in 2018
- The **Deep Web** alone is estimated to be **500 times** the size of the **surface** or indexable **Web**
 - in excess of **one billion structured data sets** as of February 2011
- It exists since Web 1.0 and it is nowadays confused with the **Dark Web**



The Dark Web

- Anonymous and confidential access to the Web and Internet
 - onion routing and strong cryptography
 - Tor (anonymous access) and I2P (anonymous hosting)
- Very often used for illegal activities

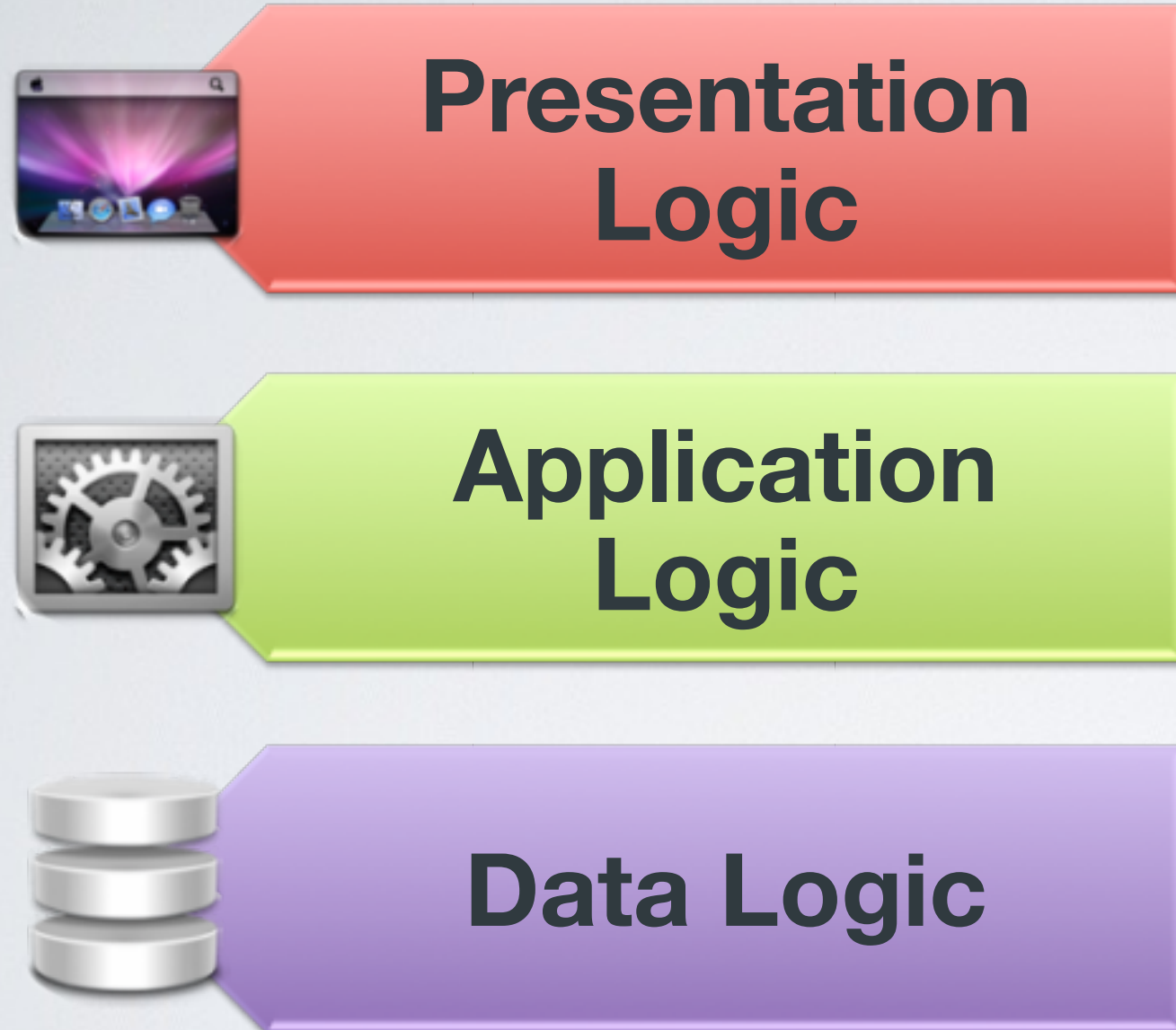




<https://www.oreilly.com/radar/why-its-too-early-to-get-excited-about-web3/>

Recap on Distributed Applications

Application Layers



● Presentation/Interface/User Logic

- manages the interaction with the user
- defines the format and visualisation of information
- accepts and validates user input

● Application/Business Logic

- defines and controls the flow of operations in the application
- defines the basic operations on the data and their constraints, often called business rules

● Data Logic

- manages the persistent storage of the data
- searches and retrieves data
- ensures consistency of the data

you do not need much cpu, mainly discs and bus

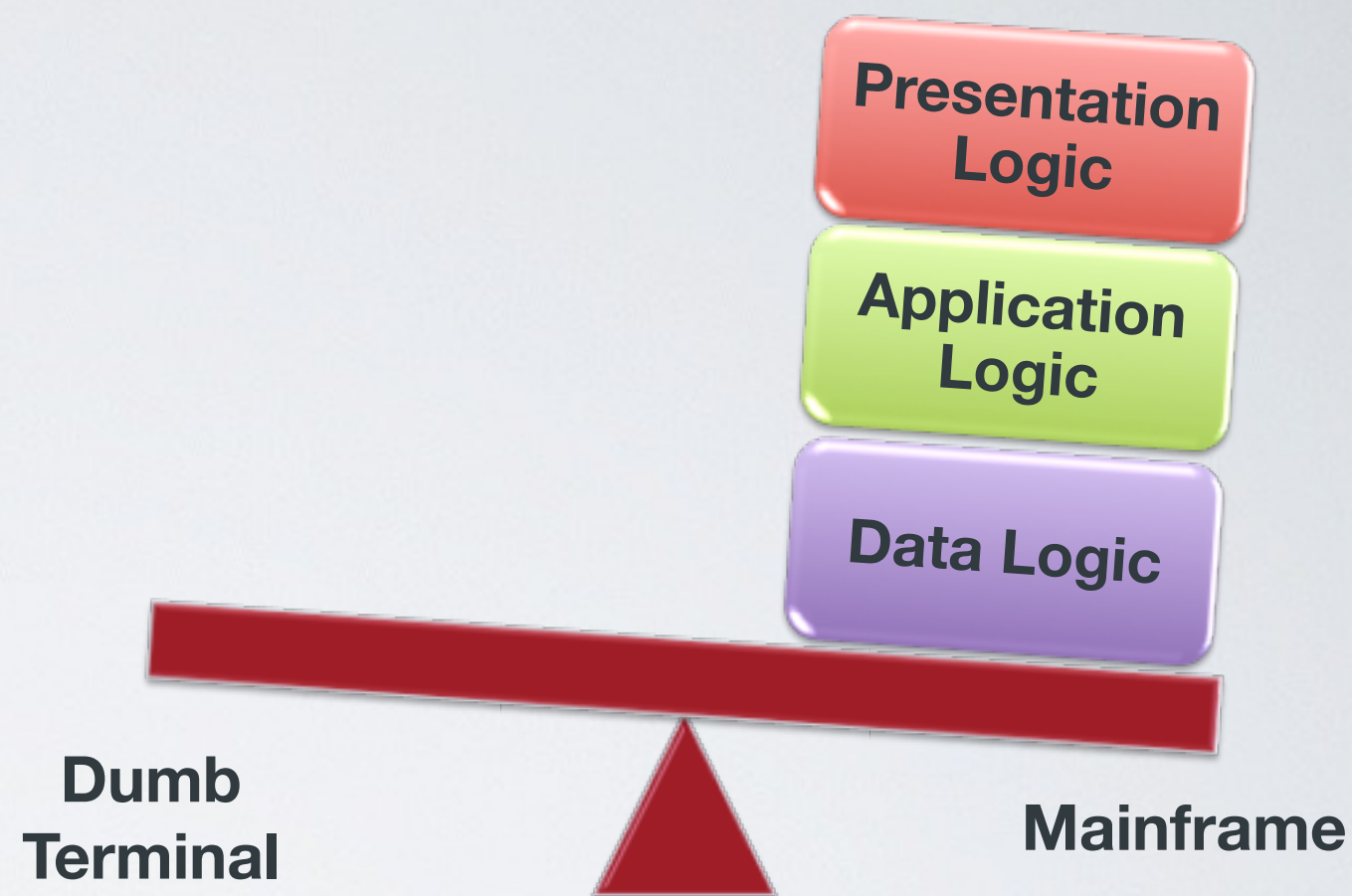
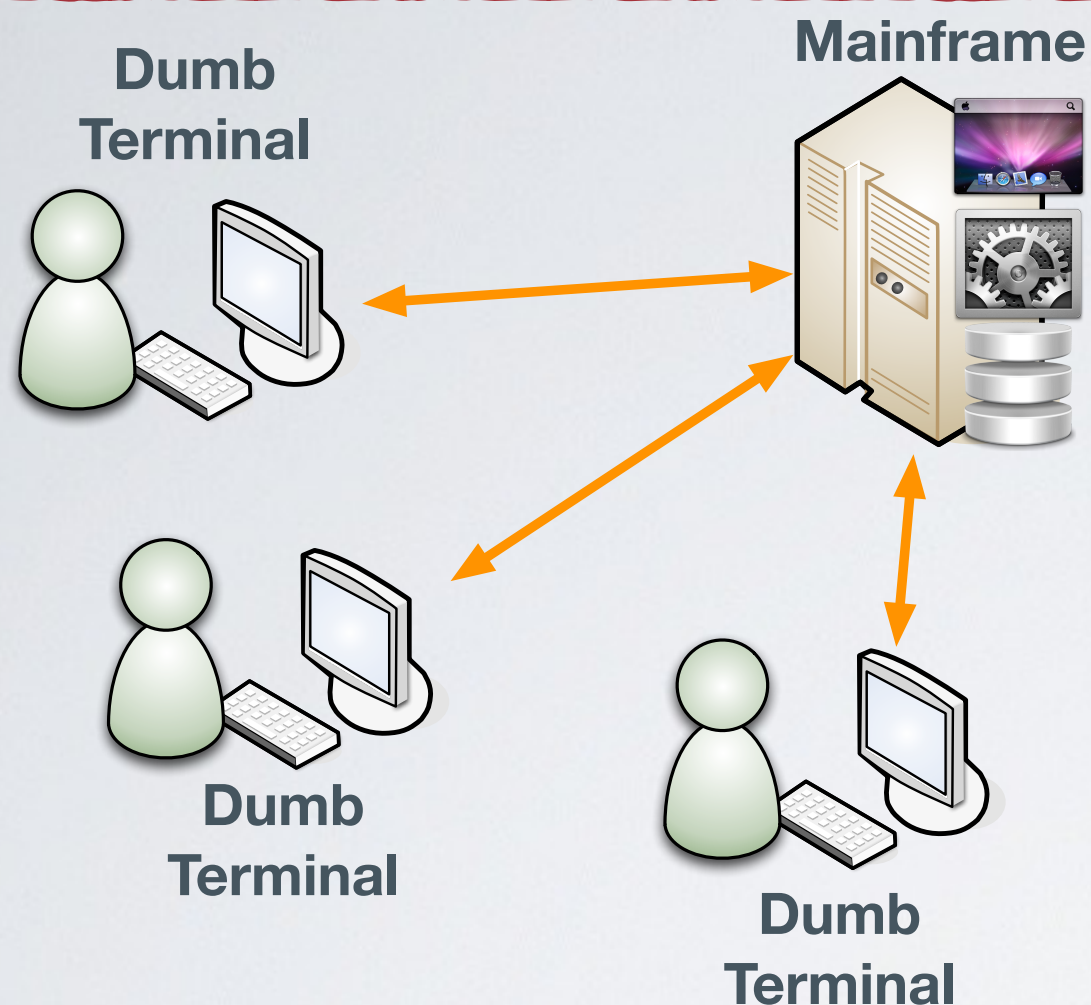
**Presentation
Logic**

**Application
Logic**

Data Logic

- Single-tier architectures
- Two-tiers architectures
- Three-tiers architectures (N-tiers)

Single-tier Architecture: Terminal/Mainframe



Pros

- easy to implement
- no management of the terminals

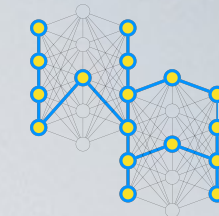
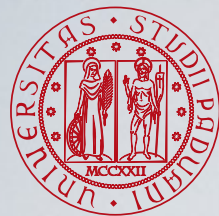
Cons

- computational load fully on the mainframe (single point-of-failure)
- scalability



The diagram shows a neural network with three layers of nodes. The top layer has 4 nodes, the middle layer has 5 nodes, and the bottom layer has 4 nodes. A path of 5 nodes is highlighted in blue, starting from the top-left node, moving down to the second layer, then right to the third layer, and finally down to the bottom layer. This path represents a specific sequence of nodes that the network might traverse during its operation.



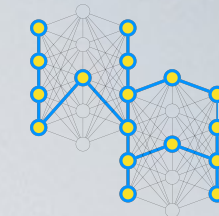
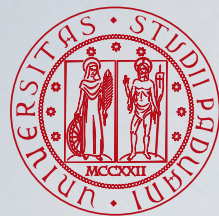


Single-tier Architecture: the example of DUO

```

VWTECSCO
File Modifica Visualizza Comunicazioni Azioni Finestra ?
+-----+
Data 30/05/05          *** DUO/2.3 ***          Ore 10:24:26.9
+-----+
| Ricerca con PAROLE su : | C          | D          | A <Periodici> |
|                          | <Cataloghi> | <Dizionario> | +-----+
| TIT Titolo EDI Editore | <alfabetici> |          | F <Funzioni> |
| AUT Autore SOG Soggetto +-----+ +-----+
| CLA Classe GEN Ovunque | B          | S          | <Lingue> | <Paesi> |
|                          | <Biblioteche> | <Soggetti> | +-----+
| <M Modulo Guidato>    | <e Sezioni> |          | <Tutti i comandi> |
+-----+
| Nome Quanti           Domande <- su> <+ giu> |
| > |
| > |
| > |
| > |
| > |
| > |
| Domanda : |
| GEN INFORMATION RETRIEVAL |
| |
| Query o Scelta M,B,C,+... | E per vedere gli <Esempi> |
+-----+
<Invio=Eseguì> <.=Menu> <Q=Fine> <I=Indietro> <H=Help> <K=Cancella>
+-----+
MA a A 20/029
Collegato con server/host remoto venere.unipd.it mediante l'utilizzo della porta 5023 HP DeskJet 500 acceso LPT1:
start Internet Explo... VWTECSCO VSTECSCO Adobe Reader - [...] Documento1 - Mic... IT 10.24

```



Single-tier Architecture: the example of DUO

```
*** DUO ***
! Domanda: GEN INFORMATION RETRIEVAL !
! Documenti trovati: 131 ( 11 pagine) !
!
! X/S      >Titoli in ordine di Data      Autore      Data !
! _ *Proceedings of SIGIR 2002 : the twenty- Beaulieu, Micheline 2002 !
! _ *String processing and information retri Laender, Alberto H.F. 2002 !
! _ *Evaluation of cross-language informatio Peters, Carol 2002 !
! _ *Mining the world wide web : an informat Chang, George 2001 !
! _ *Lectures on information retrieval : 3. Agosti, Maristella 2001 !
! _ *Principles of visual information retrie Lew, Michael S. 2001 !
! _ *Lavoro one : i quattro codici del lavor sviluppo software by S 2001 !
! _ *Proceedings of the 24. annual internati *Annual *international 2001 !
! _ *Mathematical foundations of information Dominich, Sandor 2001 !
! _ *Cross-language information retrieval an Peters, Carol 2001 !
! _ *Automatic indexing and abstracting of d Moens, Marie-Francine 2000 !
! _ *Advances in information retrieval : rec Croft, W. Bruce 2000 !
!
! Pag. 1 ( nn =numero pagina) !
! Per vedere i Titoli cliccarli o segnarli con X (invio). S per stamparli. !
!
! <V=Vederli> <F=Filtrarli> <S=Stamparli> <O=Riordinarli> <P=Periodici>
! <Invio=Avanti> <Indietro> <Top><.=Esci> <N=Nuova Ricerca> <Q=Fine Sessione>
```


Single-tier Architecture: the example of DUO

```

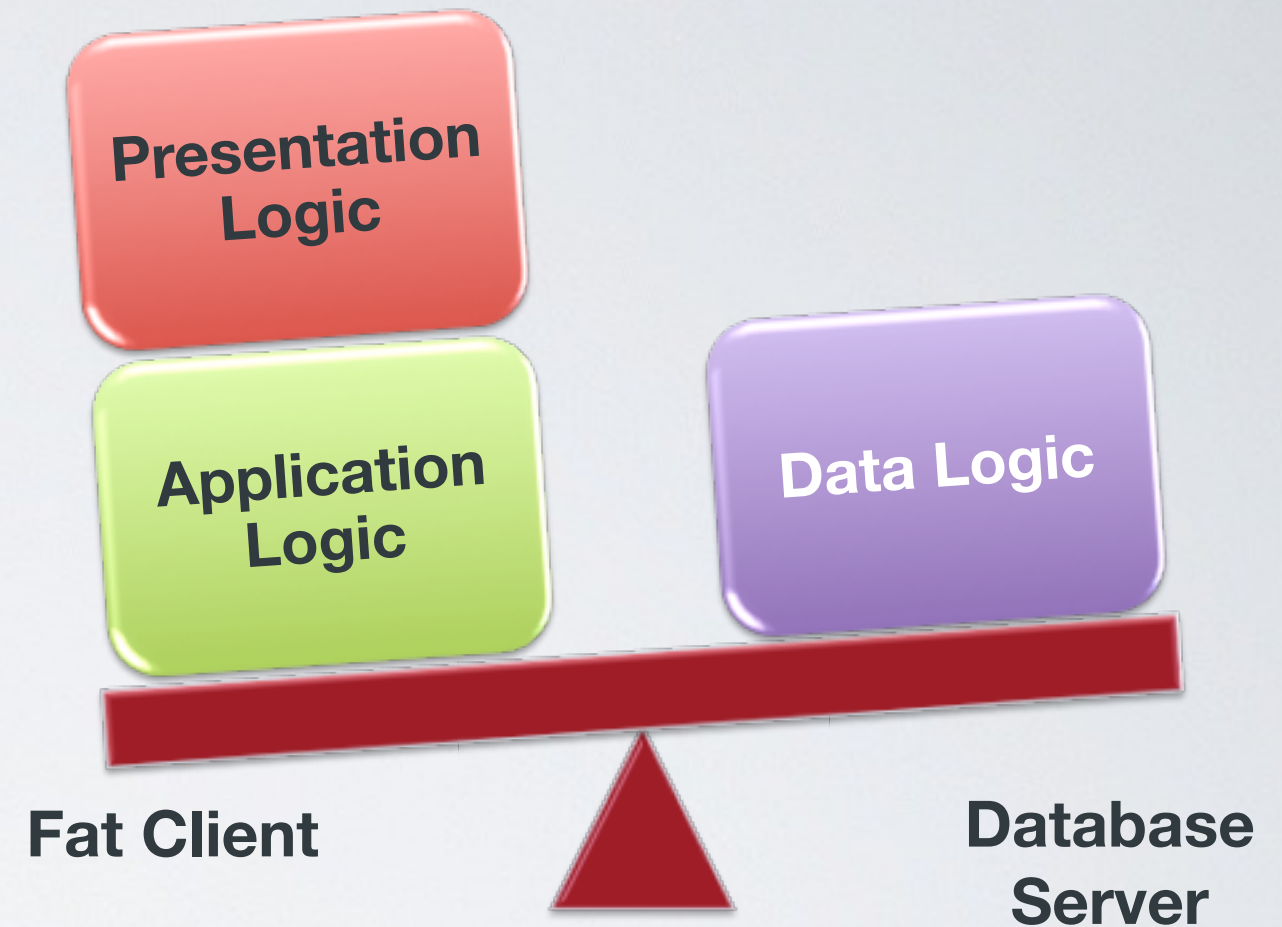
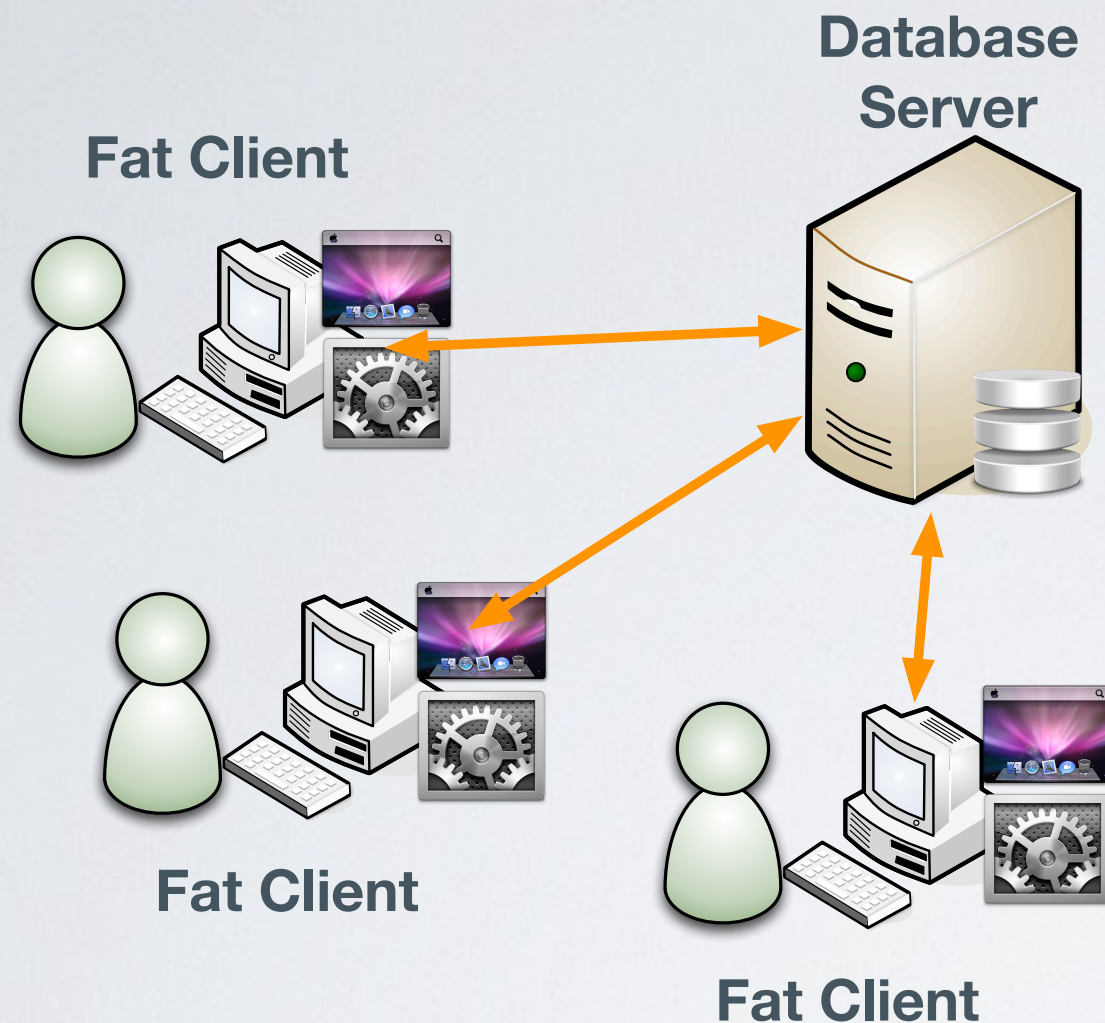
VITECD0
File Modifica Visualizza Comunicazioni Azioni Finestra ?
[Icons]

Codice titolo: PUV0857101      *** DUO ***
+-----+
Qualificazione Monografia - Lingua Inglese - Paese Stati Uniti - Data sempl
Bibliografica ice 2002 - ISBN 1581135610
-----
> Titolo      *Proceedings of SIGIR 2002 : the twenty -fifth annual
>              international ACM SIGIR conference on research and
>              development in information retrieval : August 11 -15, 2002,
>              Tampere, Finland / edited by Micheline Bealieu ... (et al.).
>              - New York : ACM press, c2002. - XVIII, 462 p. ; 28 cm.
>              ----- AUTORI -----
> Curatore   Beaulieu, Micheline
> Primo autore *Annual *international *ACM *SIGIR conference on research
>              and development in information retrieval <25. ; 2002

----- Biblioteche che possiedono questo documento -----
> Biblioteca: SC.STATISTICHE <Sezione>: I Inventario: 82716
> Segnatura:   I.25-888

Scelta: _
      <R=Dati Ridotti> <.=Esci/Prossimo> <I=Indietro> <E=Espandi> <H=Help>
Se non avete il mouse: Posizionare il cursore su una riga e premere INVIO
  
```

Two-tiers Architecture: Fat Client/Server

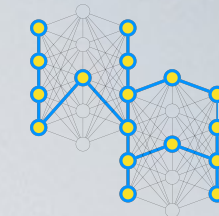
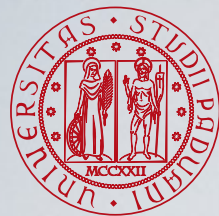


Pros

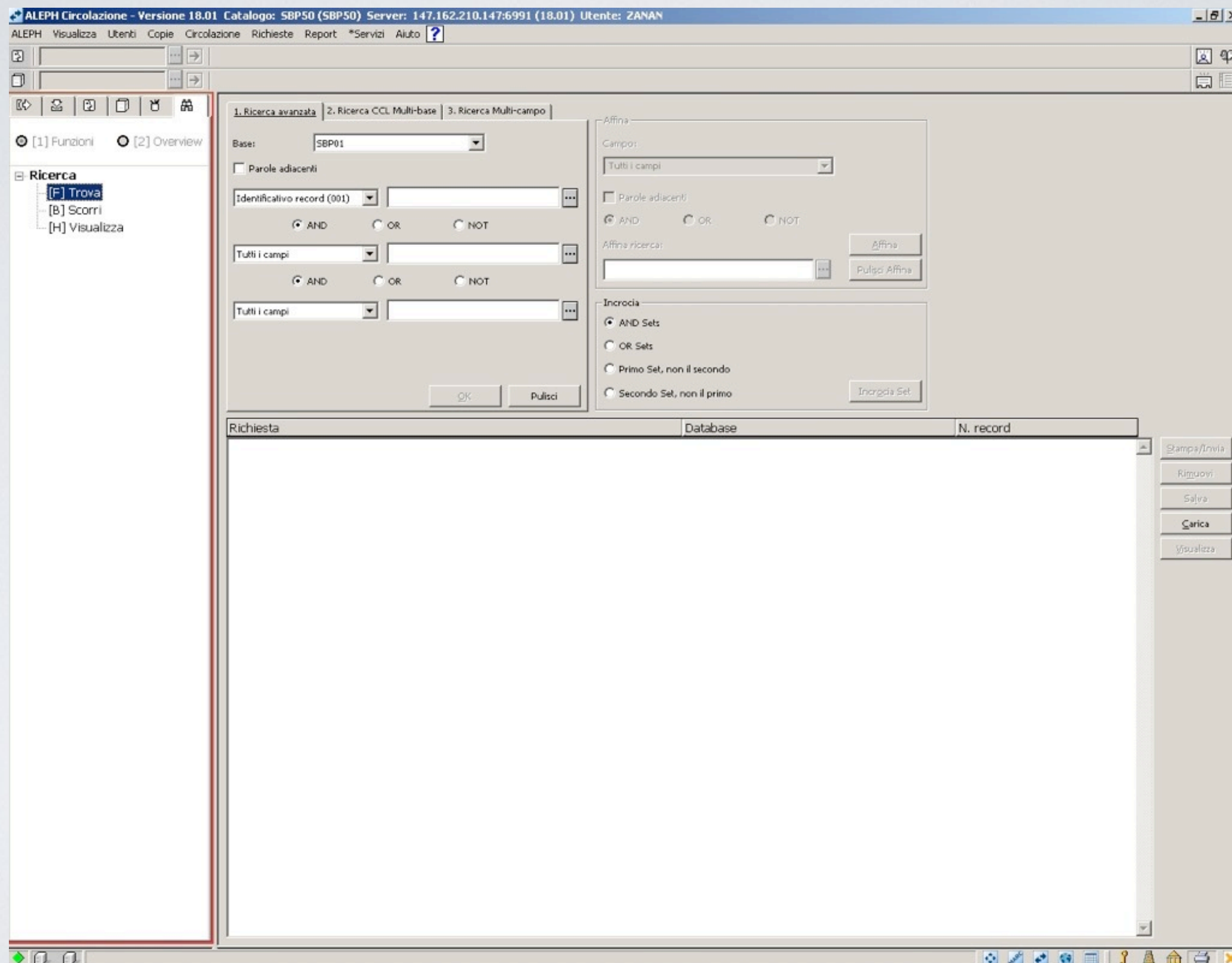
- easy to implement
- possibility to balance load

Cons

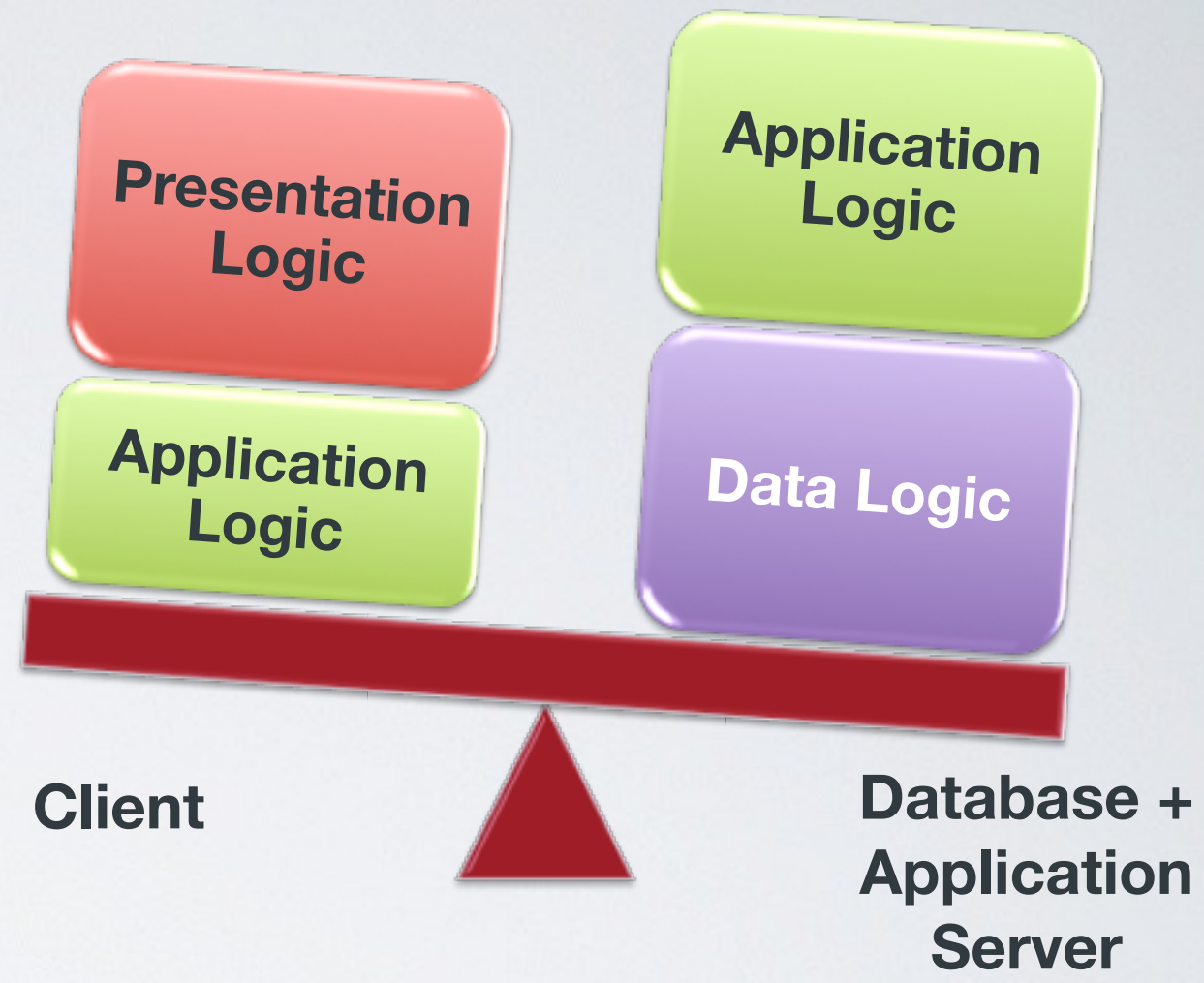
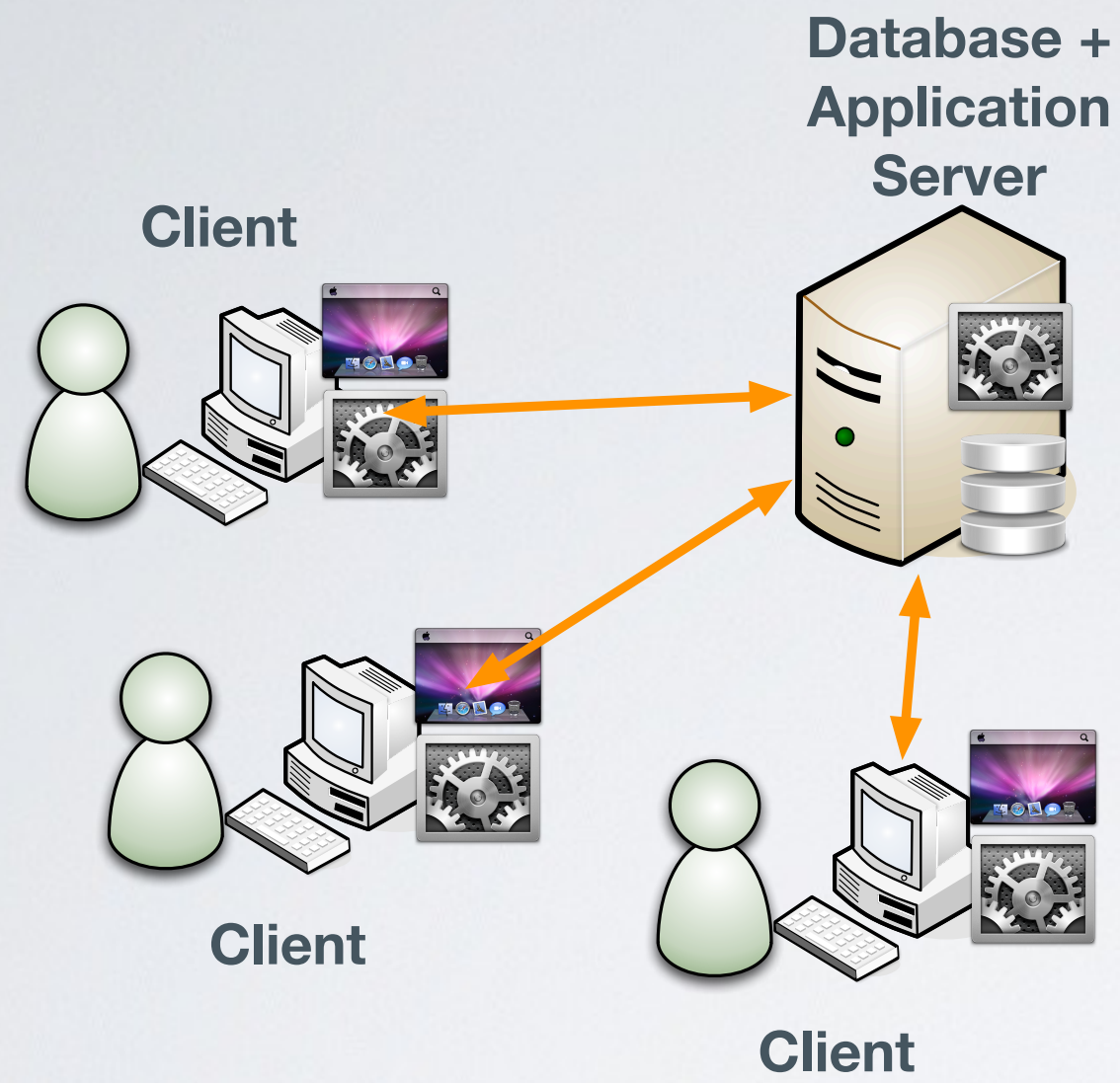
- client maintenance
- scalability



Two-tiers Architecture with Fat Client: the Example of Aleph



Two-tiers Architecture: Client/Fat Server



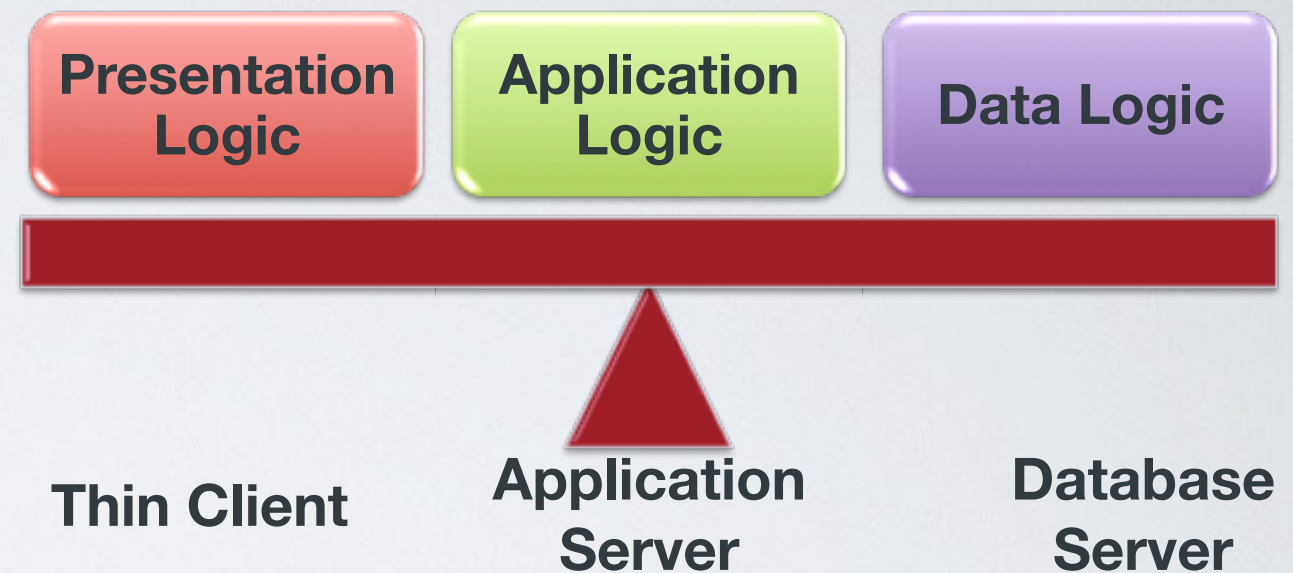
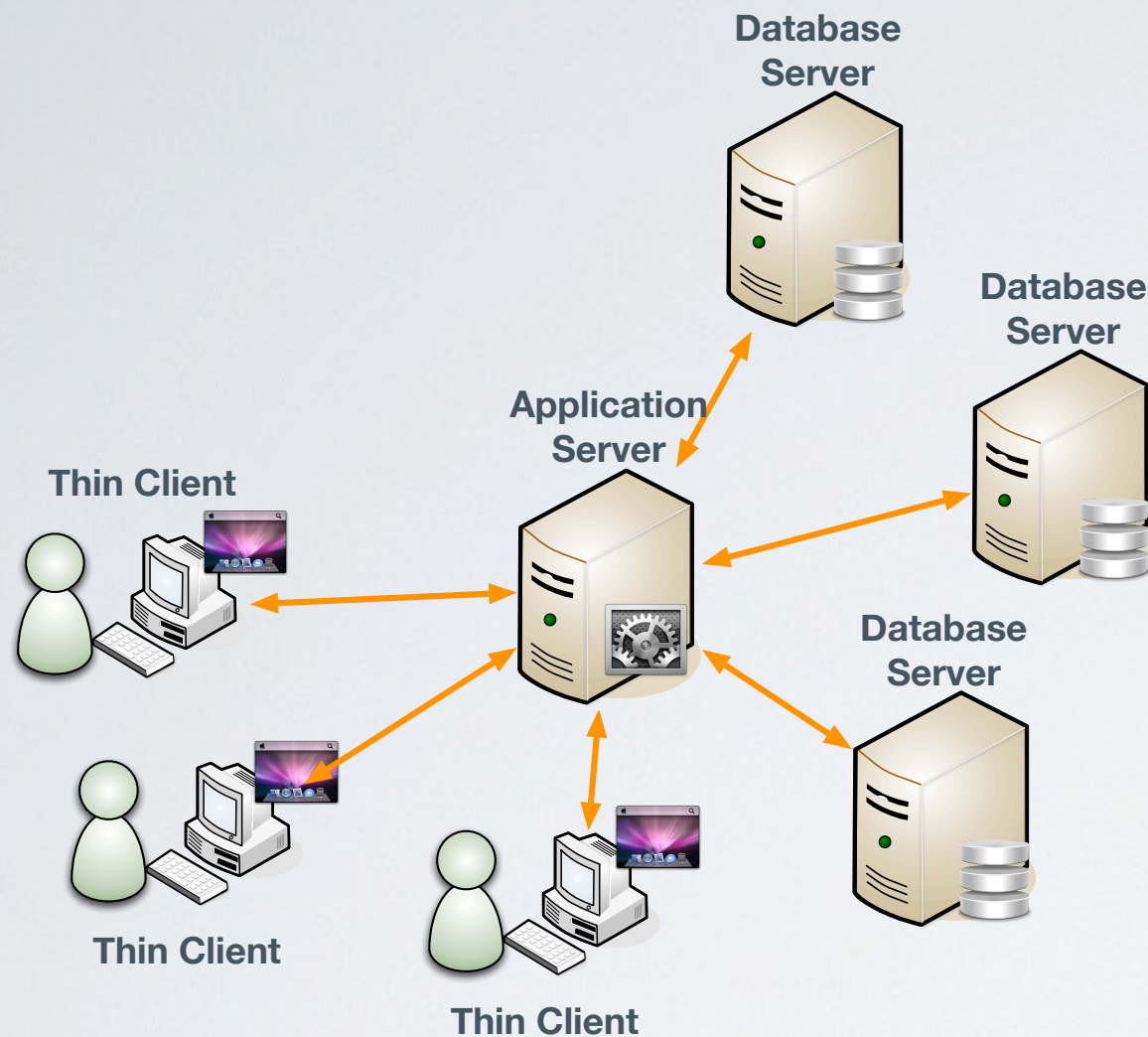
Pros

- easy to implement
- possibility to balance load

Cons

- client maintenance
- scalability

Three-tiers Architecture: Client/Middleware/Server



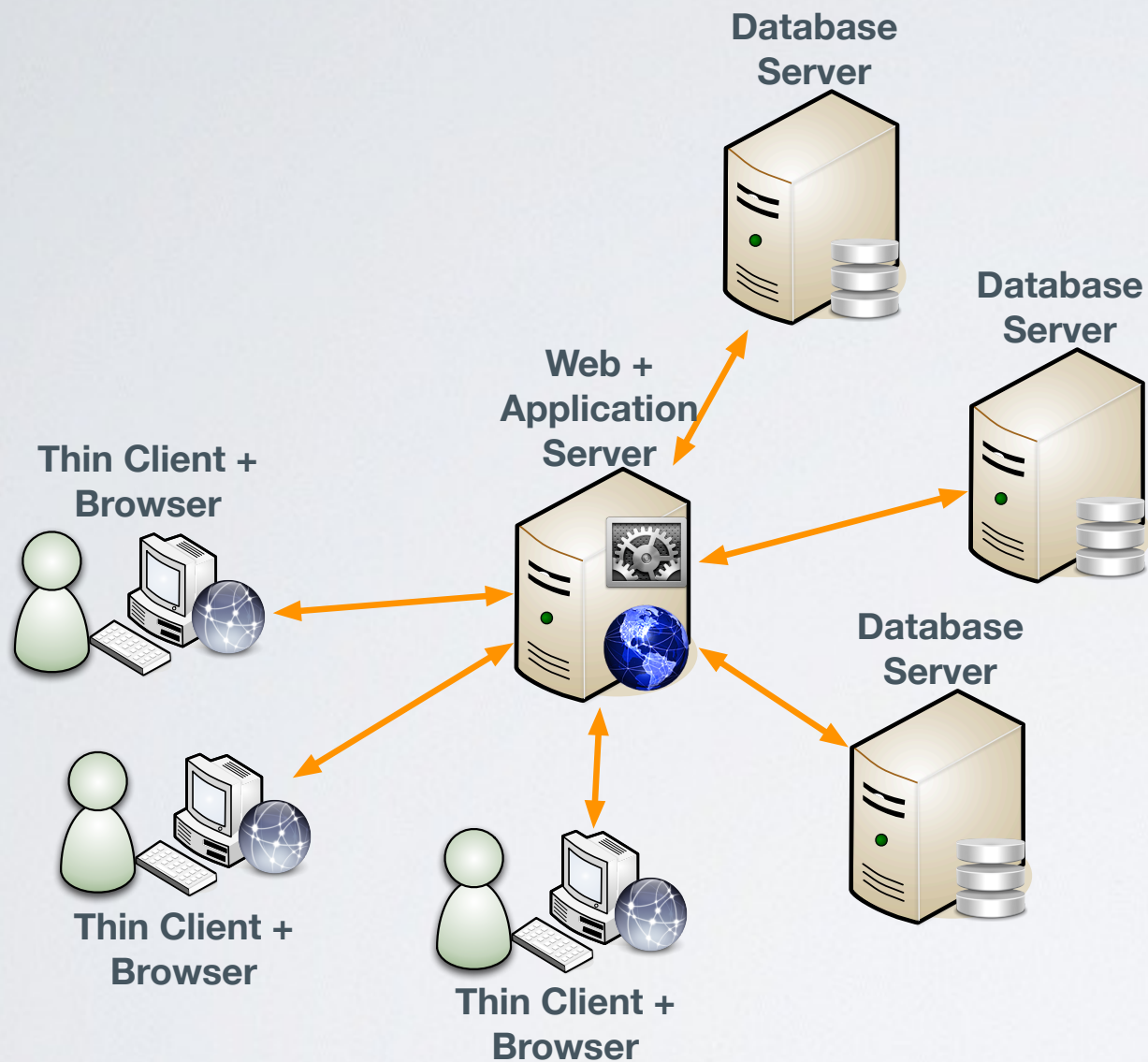
● Pros

- easy client maintenance
- possibility to balance load
- high scalability

● Cons

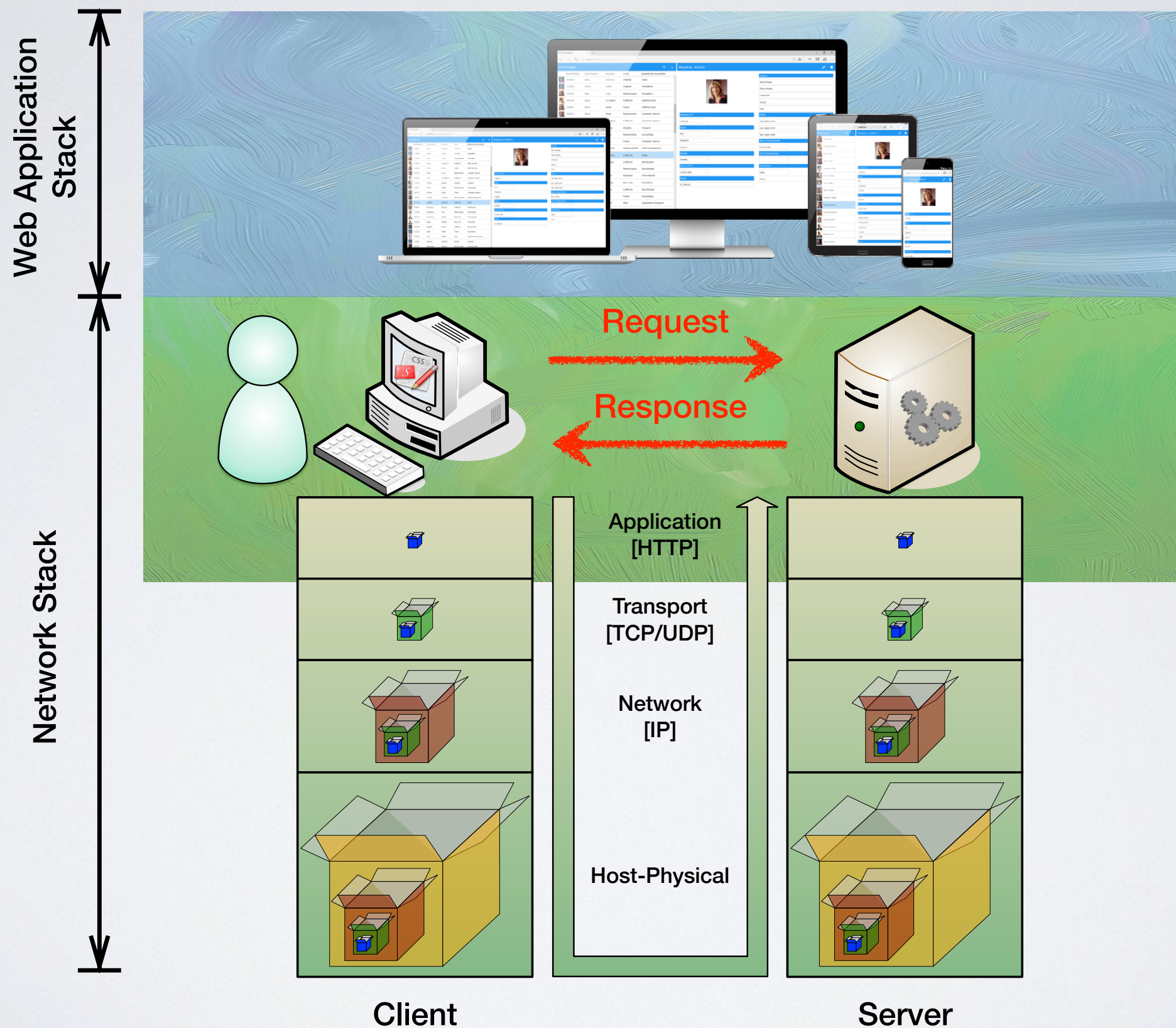
- higher implementation complexity

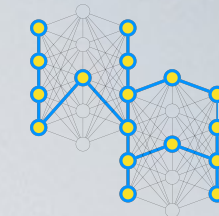
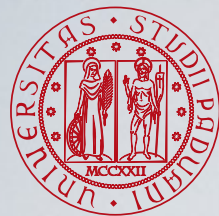
Web Applications



- They are an instance of a three-tiers architecture
- They are based on standard and ubiquitous technologies (server-side), typically already part of an organisation IT infrastructure
- They do not require any management of the clients (browser)
- Clients are ubiquitous and allow access from any kind of devices (desktop and mobile)
- Users are already comfortable with the basic interaction patterns and interface elements

Web Applications from a Computer Network Perspective





Web Applications: the Example of Aleph

[Accessibilità](#) [Aiuto](#) [Autenticati](#) [Italiano](#) [English](#)

A+AA-



UNIVERSITÀ
DEGLI STUDI
DI PADOVA



SISTEMA BIBLIOTECARIO
DI ATENEO

Biblioteca digitale

- Lo scaffale delle risorse
- Il CercaFacile
- Connessione da remoto
- Gestione bibliografie
- Collezioni digitali
- Mostre virtuali
- Archivi istituzionali
- Per chi pubblica

Cosa cerchi

- Libri e riviste
- Articoli e banche dati
- Citazioni
- Laboratori di formazione
- Prestito interbibliotecario
- Libri in formato alternativo
- Aiuto

Chi sei

- Studente
- Docente e Ricercatore
- Laureato
- Utente esterno
- Staff
- Aziende

Chi Siamo

- Il Sistema Bibliotecario di Ateneo
- Le biblioteche
- I servizi delle biblioteche
- Nuovo Archivio di Legnaro
- Il Centro di Ateneo per le Biblioteche
- Il Comitato Tecnico Scientifico
- Carta dei servizi
- Progetti

Tu sei qui: [Home](#)



La preistoria immaginata

Gli animali e i paesaggi preistorici visti attraverso gli occhi degli studiosi del XIX e XX secolo

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Ultime notizie

[1 marzo. Inaugurazione della Biblioteca di Scienze dell'Antichità Arte Musica Liviano](#)

27/02/2018

[Connessione off-campus alla Biblioteca Digitale: non solo proxy](#)

25/02/2018

[23 febbraio. Mi illumino di meno e cammino fino alla luna](#)

21/02/2018

[Altre notizie...](#)

Hai bisogno di aiuto?

- [Library System International area](#)
- [Sistema Bibliotecario di Ateneo. Infografica](#)
- [Phaidra](#)
- [Amministrazione Trasparente](#)

Libri e riviste **Citazioni** **Bibliografia** **Tesi** **Collezioni digitali**

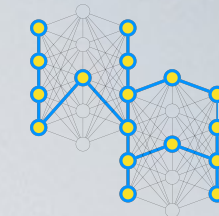
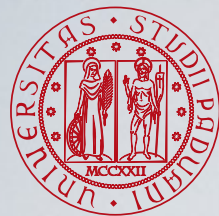
Tipo di ricerca:

Parola o frase da cercare:

Tipo di materiale:

[Per una ricerca più dettagliata vai al Catalogo](#) | [Aiuto](#)

[Cercafacile tutorial](#)



Web Applications: the Example of Aleph

CATALOGO
del sistema bibliotecario padovano

Autenticati | Fine sessione |  

Info | Aiuto

Preferenze
La mia tessera della biblioteca 

Ricerca veloce...

Ricerca semplice | Altre ricerche | Ultima ricerca | Ricerche eseguite | La mia cartella 

Catalogo > Catalogo Generale > Risultati ricerca [Cambia catalogo...](#)

Seleziona tutto
Riordina

Deseleziona
Filtra

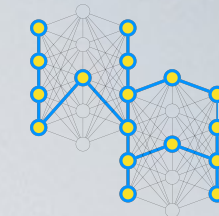
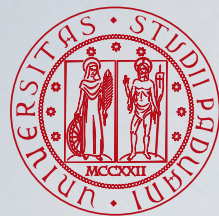
Vedi selezione
Metti in cartella

Crea sottoinsieme
Invia/Salva/Esporta

Risultati per Parole= information retrieval; Ordinati per: Anno (discendente)/Autore [Salva la ricerca...](#)
Opzioni di ordinamento: ▶ Autore/Titolo ▶ Autore/Anno ▶ Titolo/Anno ▶ Anno/Autore
Visualizza: ▶ Autore-Titolo-Copie ▶ Titolo-Copie ▶ Formato Breve

Record: 101– 120 su 264 - Sono visualizzati e ordinati al massimo 1000 record
 1 2 3 4 5 6 7 8 9 10 11 12 13 14

Nr.	Autore ▼	Formato	Titolo ▼	Anno	Biblioteca/Copie	Legame esterno
101	☐ Stacey, Alison	 Libro	Effective information retrieval from the Internet / Alison Stacey and Adrian Stacey	 2004	Mediateca Centro Linguistico(1/ 0) Polo di Lettere(1/ 0)	
102	☐	 Libro	Multimedia information retrieval / a cura di Roberto Raieli e Perla Innocenti ; premessa di Luci	 2004	Mediateca Centro Linguistico(1/ 0) Polo Beato Pellegrino(1/ 0) Polo di Lettere(3/ 0)	
103	☐ Chakrabarti, Soumen	 Libro	Mining the web / Soumen Chakrabarti	 2003	Polo di Ingegneria(1/ 0) Polo di Scienze(2/ 0) Polo di Scienze Sociali(1/ 0)	
104	☐ Chu, Heting	 Libro	Information representation and retrieval in the digital age / Heting Chu	 2003	Polo di Lettere(1/ 0)	
105	☐ Croft, W. Bruce	 Libro	Language modeling for information retrieval / edited by W. Bruce Croft and John Lafferty	 2003	Polo di Scienze Sociali(1/ 0)	
106	☐ Ferrero, Marco	 Libro	Laboratorio Access 2002 / Marco Ferrero	 2003	Biblioteca Chiereghin Treviso(1/ 0) Polo di Scienze(1/ 0)	
107	☐ Allan, James	 Libro	Topic detection and tracking / edited by James Allan	 2002	Polo di Scienze Sociali(1/ 0)	
108	☒ Annual international ACM SIGIR conference on research and development in information retrieval <2	 Libro	Proceedings of SIGIR 2002 / edited by Micheline Beaulieu ... \et al.!	 2002	Polo di Scienze Sociali(1/ 0)	
109	☐ Colomb, Robert M.	 Libro	Information spaces / Robert M. Colomb	 2002	Polo di Scienze(	



Web Applications: the Example of Aleph

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La mia tessera della biblioteca 

Ricerca veloce...
Vai

Info | Aiuto

Ricerca semplice Altre ricerche Ultima ricerca Ricerche eseguite La mia cartella

Catalogo > Catalogo Generale > Risultati ricerca > Formato completo del record Cambia catalogo...

Metti in cartella Invia/Salva/Esporta 

Cambia formato: ▶ Standard ▶ Scheda catalografica ▶ Citazione ▶ Nomi dei campi ▶ Campi MARC

Record 108 di 264  Link permanente

Cerca in Google

Amazon.com

libreriauniversitaria.it

WorldCat Identities

WorldCat Identities

Titolo

Pubblicazione

Descr.

Autore

Altro autore

ISBN

BID

Lingua

Tipo di Materiale

Dove?

▶ Proceedings of SIGIR 2002 : the twenty-fifth annual international ACM SIGIR conference on research and development in **information retrieval** : August 11-15, 2002, Tampere, Finland / edited by Micheline Beaulieu ... et al.!

▶ New York : ACM press, c2002

XVIII, 462 p. ; 28 cm.

▶ Annual international ACM SIGIR conference on research and development in **information retrieval** <25. ; 2002 ; Tampere>

▶ Beaulieu, Micheline

1581135610

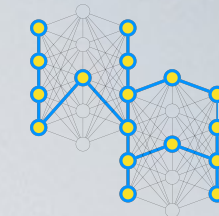
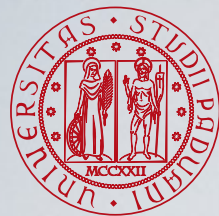
PUV0857101

Inglese

Libro

Mostra le biblioteche

Cambia formato: ▶ Standard ▶ Scheda catalografica ▶ Citazione ▶ Nomi dei campi ▶ Campi MARC



Web Applications: the Example of Aleph

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sba

Catalogo > Catalogo Generale > Risultati ricerca > Formato completo del record > Posseduto

Indietro

Proceedings of SIGIR 2002 : the twenty-fifth annual international ACM SIGIR conference on research and development in information retrieval : August 11-15, 2002, Tampere, Finland / edited by Micheline Bealieu ... [et al.]

✓ Clicca su "Prenota" per prenotare il documento, su "Espandi" per vedere il record completo

✓ Riviste: Clicca su "Ultimi fascicoli" per vedere i fascicoli dell'ultimo anno (funzione non sempre disponibile)

Mostra solo:

Anno Tutti

Volume Tutti

Biblioteca Tutte

☐ Nascondi libri a prestito

Vai

◀ Precedente

Successiva ▶

Prestito	Descrizione	Polo/Biblioteca	Collocazione	Status	Disponibilità *	Prenotazioni	AIRE
Espandi		Polo di Scienze Sociali	Biblioteca di Scienze Statistiche 'Bernardo Colombo' I.25-888	Prestabile	A scaffale		

◀ Precedente

Successiva ▶

* Quando compare una **Data**: il libro è a prestito fino a tale data.
Per prenotare titoli del **Deposito Legnaro** e **In Deposito** clicca su "Prenota"; se non compare il link "Prenota", rivolgiti alla tua biblioteca.

Web Applications, A.Y. 2023/2024
MD in “Computer Engineering”, “Cybersecurity”, and “ICT for Internet and Multimedia”

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Questions?

